

Cut through traffic like a snake: cooperative adaptive cruise control with successive platoon lane-change capability

Haoran Wang, Xin Li, Xianhong Zhang, Jia Hu, Xuerun Yan & Yongwei Feng

To cite this article: Haoran Wang, Xin Li, Xianhong Zhang, Jia Hu, Xuerun Yan & Yongwei Feng (2024) Cut through traffic like a snake: cooperative adaptive cruise control with successive platoon lane-change capability, *Journal of Intelligent Transportation Systems*, 28:2, 141-162, DOI: [10.1080/15472450.2022.2114081](https://doi.org/10.1080/15472450.2022.2114081)

To link to this article: <https://doi.org/10.1080/15472450.2022.2114081>



Published online: 24 Aug 2022.



Submit your article to this journal [↗](#)



Article views: 798



View related articles [↗](#)



View Crossmark data [↗](#)



Citing articles: 21 View citing articles [↗](#)



Cut through traffic like a snake: cooperative adaptive cruise control with successive platoon lane-change capability

Haoran Wang^a, Xin Li^b, Xianhong Zhang^c, Jia Hu^a, Xuerun Yan^a, and Yongwei Feng^a

^aKey Laboratory of Road and Traffic Engineering of the Ministry of Education, Tongji University, Shanghai, China; ^bCollege of Transportation Engineering, Dalian Maritime University, Dalian, China; ^cAutonomous Driving Business Unit, SAIC Motor Corporation Limited, Shanghai, China

ABSTRACT

A cooperative adaptive cruise control (CACC) system may be impeded by slow-moving traffic in the application. To improve the mobility of CACC, this research proposes a CACC controller with successive platoon lane-change capability. The goal is to help a platoon cut through traffic successively like a snake via smaller windows. The proposed controller has the following features: i) with successive platoon lane-change capability; ii) with string stability and lateral stability; iii) with consideration of vehicle dynamics. The proposed controller is evaluated on a simulation platform with the context of traffic and a joint simulation platform consisting of PreScan and Matlab/Simulink. The Results demonstrate that compared to the conventional controller: i) platoon lane-change competence is enhanced by 71.36% on arterials and 120.49% on freeways; ii) platoon lane-change efficiency is enhanced by 25.05% on arterials and 41.36% on freeways; iii) the proposed controller is more robust against congestion. Moreover, the computation time of the proposed controller is approximately 15 milliseconds when running on a laptop equipped with an Intel i7-8750H CPU. This indicates that the proposed controller is ready for real-time implementation.

ARTICLE HISTORY

Received 14 February 2022

Revised 2 August 2022

Accepted 14 August 2022

KEYWORDS

CACC; motion planning;
platoon lane change;
space-domain;
stability analysis

Introduction

Improving mobility, sustainability, and safety are three major goals of transportation operations (Hu et al., 2022). However, 81.3% of cities around the world are experiencing an increasing level of traffic congestion according to TomTom Traffic Index 2016. Traffic congestion not only deteriorates transportation mobility (Treiterer & Myers, 1974) but also drives up fuel consumption (Qin et al., 2021) and emissions (Barth & Boriboonsomsin, 2008). Moreover, in 2017, 37,133 traffic fatalities occurred in the US. It indicates that 1.16 fatalities occurred per 100 million vehicle miles traveled (National Center for Statistics and Analysis, 2019, June).

The emerging Cooperative Adaptive Cruise Control (CACC) system is able to mitigate the aforementioned problems (Bibeka et al., 2021; Liu et al., 2021; Qin & Wang, 2021; Qin et al., 2021). CACC forms Connected Automated Vehicles (CAVs) into a platoon. The platoon time headway can drop to as low as 0.5 s according to field tests, much smaller compared to the 1–2 s smallest time headway of human-

driven vehicles (HVs) (Shladover et al., 2012). This time headway reduction leads to an increase in traffic capacity to over 4,200 veh/h/ln, which nearly doubles conventional road capacity (Vander Werf et al., 2002). Cooperation among vehicles is also expected to reduce 12% fuel consumption and 14% carbon dioxide emission (Park et al., 2011). In addition, CACC is forecasted to improve traffic safety by reducing the diversity of heterogeneous driving behavior, which is regarded as one of the major causes of crashes (Dey et al., 2016).

When CACC is implemented in the field, the lateral lane-change capability is critical since a platoon needs to change lane from time to time. Lack of lane-change capability forces a CACC platoon to disengage when lane change cannot be avoided (Bengtsson et al., 2015; Liu et al., 2018; Toy et al., 2002; Weaver et al., 2021). This platoon disengagement not only calls for unnecessarily additional human take-over maneuvers but also breaks the running continuity of CACC systems (Hu et al., 2020).

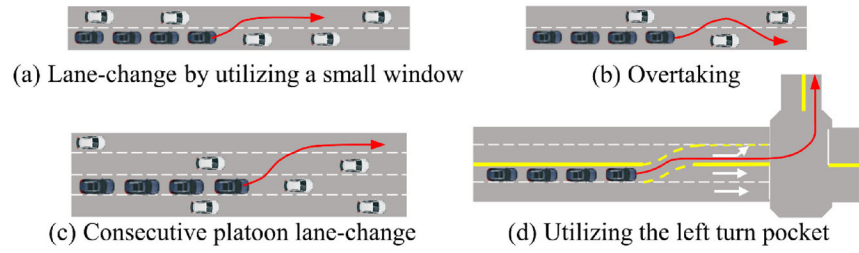


Figure 1. Application examples of SuPLC.

However, only a few past studies make efforts on developing platoon lane-change (PLC) controllers for CACC. One of the conventional PLC methods is to disengage the platoon and control vehicles to change lane freely (Hsu & Liu, 2004; Lam & Katupitiya, 2013). This method breaks the platoon form and control continuity. Hsu and Liu proposed a PLC controller to keep the form of a platoon when changing lane (Hsu & Liu, 2008). Whereas, this PLC controller plans lateral motions by computing lateral acceleration without considering vehicle dynamics. Therefore, the planned path may not be feasible for a vehicle to fulfill. This problem could be handled by an artificial potential field based controller that considers vehicle dynamics (Huang et al., 2019). However, this controller may result in a local optimum that leads to lateral oscillations and along with it – risks. Moreover, most PLC controllers only allow CAVs in a platoon to change lane simultaneously. The capability of the simultaneous platoon lane-change (SiPLC) method highly relies on the chance of finding a big-enough gap. For a CACC system with five CAVs, one-second headway, and 65 mph cruising speed, the platoon length is about 150 meters. In order to maneuver the platoon to change lane simultaneously, a gap greater than 150 meters is needed. This gap is in correspondence with the traffic density rated at the level of service A. Therefore, the SiPLC method significantly reduces the chance to conduct lane change and further damages mobility. However, in scenarios shown in Figure 1, lane-change windows are not enough for a SiPLC maneuver, while PLC is still needed. Hence, a next-generation successive platoon lane change (SuPLC) is proposed in this paper to increase mobility and ensure the running continuity of CACC. The SuPLC controller has a different logic compared to the conventional SiPLC controller, as shown in Figure 2. It enables a platoon to change lane in turn by adopting smaller lane-change gaps. Therefore, it increases the lane-change capability of CACC.

Platoon stability is necessary for safeguarding against disturbances in the field application of CACC (Ploeg et al., 2011). A variety of stabilities shall be

ensured, such as local stability (Zhang et al., 2021), string stability (Khound et al., 2021; Ploeg et al., 2014; Seiler et al., 2004; Swaroop & Hedrick, 1996), stability margin (Hao et al., 2011), and behavior coherence (Bamieh et al., 2012). Past studies have developed various CACC controllers with stability under its regular cruise status. However, CACC controllers with stability under lane-change status have not yet been developed. Moreover, the SuPLC maneuver is longitudinal and lateral coupled. Followers track both longitudinal and lateral motions of the leader. Therefore, both longitudinal and lateral stability of a string should be proven. Lateral stability has only been proven for single-vehicle control (Li et al., 2017; Xu et al., 2014). The concept of string lateral stability has not been mentioned in past studies.

In this paper, a next-generation CACC controller is proposed with enhanced lane-change capability. The proposed controller is with the following features:

- With successive platoon lane-change capability;
- Ensuring string and lateral stability;
- With consideration of vehicle dynamics.

The remainder of this paper is organized as follows. Section “Research scope” presents the research scope of this paper. Section “Problem formulation” formulates the SuPLC controller and presents a solution algorithm. Section “Performance analysis” analyses the performance of the proposed controller and proves its stability. Section Evaluations evaluates the proposed controller in simulation. Section “Conclusion and future research” makes a conclusion and provides future research ideas.

Research scope

The goal of the proposed controller is to maneuver a CACC platoon to cut through traffic successively like a snake. There are three highlights of this controller:

- **Smaller lane-change window requirement:** The proposed controller can control CAVs to change

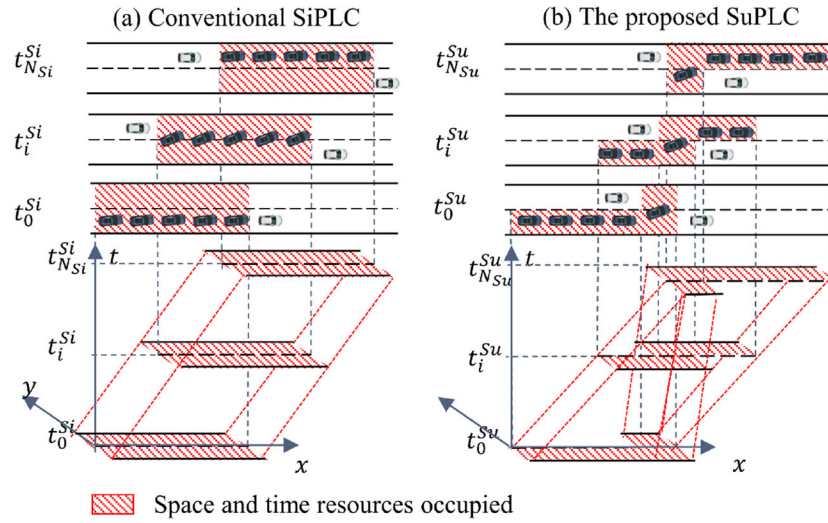


Figure 2. Time-space resources allocation comparison between SiPLC and SuPLC.

lane in turn. Therefore, a moving lane-change window that is large enough for a single CAV's lane-change maneuver is also theoretically sufficient for platoon lane-change maneuver. Lower platoon lane-change window threshold provides platoons a better chance to cut through traffic. It further improves the platoon lane-change capability. This highlight is demonstrated in the evaluation.

- **Reducing needs for platoon disengagement:** In scenarios such as utilizing a left turn pocket or exit ramp, the platoon has to be disengaged under the control of conventional controllers. However, the proposed SuPLC controller can keep running continuity in these scenarios.
- **String and lateral stable:** The proposed controller is proven with string and lateral stability.

System structure

The proposed SuPLC controller is adopted in the CACC system as presented in Figure 3. The system consists of three modules. Module 1 makes platoon lane-change decisions. Module 2 plans trajectories and motions for the SuPLC maneuver. Module 3 is an execution layer. The research scope of this paper is module 2 – the planning of the SuPLC maneuver.

Details of the system are provided as follows:

- **Module 1:** This module makes decisions on platoon lane change. First, a choice shall be made between SiPLC and SuPLC based on real-time traffic statuses. The SuPLC method is adopted when there is not enough SiPLC space. Then, a PLC window is selected for the SuPLC maneuver. This paper proposes a simple rule-based platoon lane-

change window selection algorithm for the evaluation with traffic, as shown in Table 1. The algorithm ensures that a SuPLC decision is made only when enough space could be found to settle down the platoon. Conservative parameters are adopted to avoid a SuPLC maneuver being cutoff.

- **Module 2:** After receiving SuPLC command and status information passed on from module 1, the platoon leader is controlled by an automated lane-change (ALC) controller (Nilsson et al., 2015) and an adaptive cruise control (ACC) controller (Kesting et al., 2008). The longitudinal control objective of the leader is to maintain the desired speed. The lateral control objective of the leader is to change lane through the selected window. All followers are controlled by the proposed SuPLC controller with the objective of following the trajectory of the leader. Control commands are passed on to module 3.
- **Module 3:** This module actuates control commands provided by module 2.

Problem statement

To achieve a successive platoon lane-change maneuver, followers shall track longitudinal and lateral motions of the leader. Therefore, the objective of the proposed SuPLC controller is decomposed into a longitudinal objective and a lateral objective as follows:

- **Longitudinal objective:** Longitudinal objective of the proposed controller is to maintain a constant and stable time headway between adjacent CAVs in a platoon.

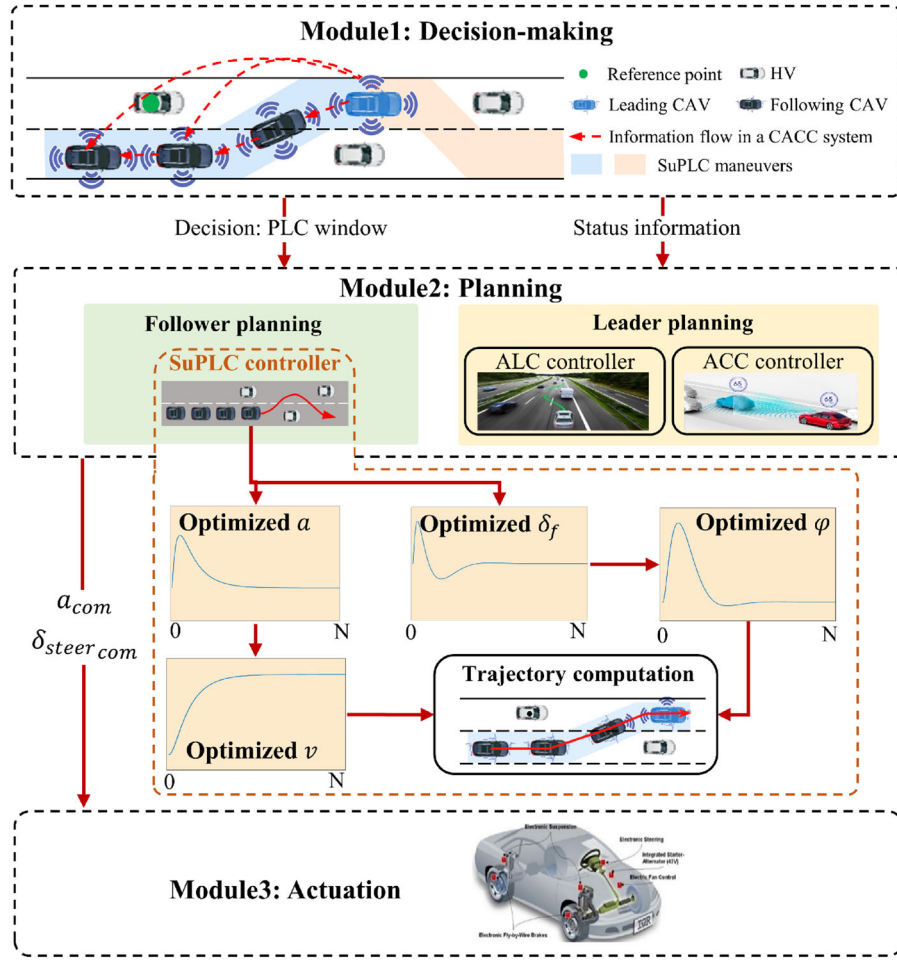


Figure 3. System logic.

- **Lateral objective:** Lateral objective of the proposed controller is to minimize lateral tracking error. Lateral tracking error e_y^j is defined as follows:

$$e_y^j(s_r) \triangleq |y^j(s_r) - y^0(s_r)| \quad (1)$$

where s_r is the longitudinal position in a relative frenet coordinate as shown in Figure 4. In this relative Frenet coordinate system, the origin is on the center-line of the road. $y^j(s_r)$ is the lateral position of the j th follower at a longitudinal position s_r ; $j = 0$ denotes the leading CAV.

Problem formulation

The formulation of the optimal control based SuPLC controller is presented in this section. It is designed to minimize longitudinal and lateral following error while fulfilling the ultimate goal: capable of maneuvering through smaller windows.

Figure 4 presents a typical scenario where a platoon makes a successive lane change.

State definition

Since the lateral objective is to minimize lateral tracking error, a space-domain-based controller can improve control accuracy compared to a time-domain-based controller. Therefore, state and control variables are defined in space-domain, as shown in Definition 1.

Definition 1 (State and control vector): The system state vector ξ and control vector u are defined as follows:

$$\xi \triangleq [\pi(s_r), y(s_r), e_\phi(s_r), t(s_r)]^T \quad (2)$$

$$u \triangleq [a(s_r), \delta_f(s_r)]^T \quad (3)$$

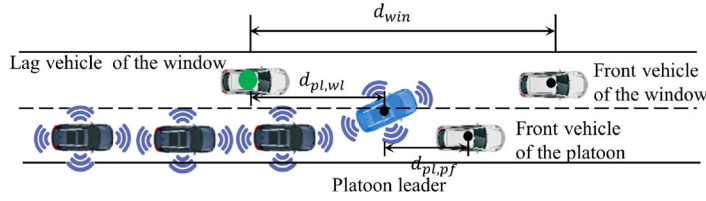
where s_r is longitudinal position; y is lateral position; e_ϕ is heading angle error; t is global time; a is acceleration; δ_f is front-wheel angle; π is defined as follows:

$$\pi \triangleq \frac{1}{v} \quad (4)$$

The desired state of ego vehicle is the state of the leader at the same longitudinal position, as presented

Table 1. Platoon lane-change window selection algorithm.

Scenario notations:

Initialize: $d_{l,min}^{LC}$; $d_{f,min}^{LC}$; h_{min}^{tra} Input: v_{des}^p ; l_{pl} ; $\mathbb{W}$; for each window: d_{win} , $d_{pl,wl}$, and $d_{pl,pf}$

Output: PLC window selection

for all windows in $\mathbb{W}$ from near to far

$$d_{l,min}^{LC} = (t_{react}^{HV} + t_{brake, delay}) \times v^{lw} - \frac{v^{lw^2}}{2a_{min}} + \frac{v^2}{2a_{min}}$$

$$d_{f,min}^{LC} = (t_{react}^{CAV} + t_{brake, delay}) \times v - \frac{v^2}{2a_{min}} + \frac{v_p^2}{2a_{min}}$$

if $d_{pl,wl} \geq d_{l,min}^{LC}$ thenif $d_{pl,pf} \geq d_{f,min}^{LC}$ thenif $d_{win} \geq d_{l,min}^{LC} + 2h_{min}^{tra}v_{des}^p + l_{pl}$ then

The window is selected for SuPLC.

where $d_{l,min}^{LC}$ is the safe lane-change window between a vehicle and its lag vehicle; $d_{f,min}^{LC}$ is the safe lane-change window between a vehicle and its front vehicle; t_{react}^{HV} is the human reaction time; t_{react}^{CAV} is the CAV reaction time; $t_{brake, delay}$ is the delay of brake; v^{lw} is the speed of the lag vehicle of the window; v_p is the speed of the front vehicle of the platoon; h_{min}^{tra} is the minimum car-following time in traffic; v_{des}^p is platoon desired speed; l_{pl} is the length of platoon; $\mathbb{W}$ is the set of alternative PLC windows.

in Definition 2. The longitudinal position could be absolute or relative in regard to the earth, depending on how the coordinate system is defined.

Definition 2 (Desired state): The ego CAV is the j th follower in the platoon. The desired state ξ_{des} is defined as follows:

$$\xi_{des} \triangleq [\pi^0(s_r), y^0(s_r), e_\phi^0(s_r), t_{des}(s_r)]^T \quad (5)$$

$$t_{des}(s_r) \triangleq t^0(s_r) + jh_{des}^p \quad (6)$$

where 0 is the index of the leader; $t_{des}(s_r)$ is the desired arrival time at a position s_r ; h_{des}^p is the desired headway in a platoon.

System dynamics

The dynamics of a CAV are formulated based on a bicycle model. The conventional bicycle model is formulated as follows (Campion et al., 1996; Dang et al., 2020; Wang et al., 2019; Zhang et al., 2019):

$$\frac{ds_r}{dt} = v \quad (7)$$

$$\frac{dy}{dt} = ve_\phi \quad (8)$$

$$\frac{de_\phi}{dt} = v \left(\frac{\delta_f}{l_f} - \kappa \right) \quad (9)$$

$$\frac{dv}{dt} = a \quad (10)$$

where δ_f is front-wheel angle; e_ϕ is heading angle; l_f is the distance between the front axle and rear axle; κ is road curvature.

By substituting equations (4) and (7) into equations (8), (9), and (10), a space-domain vehicle dynamics model is given as follows:

$$\frac{dt}{ds_r} = \pi \quad (11)$$

$$\frac{d\pi}{ds_r} = -\pi^3 a \quad (12)$$

$$\frac{dy}{ds_r} = e_\phi \quad (13)$$

$$\frac{de_\phi}{ds_r} = \frac{\delta_f}{l_f} - \kappa \quad (14)$$

By applying the Euler Method and Definition 1, equations (11), (12), (13), and (14) are discretized as follows:

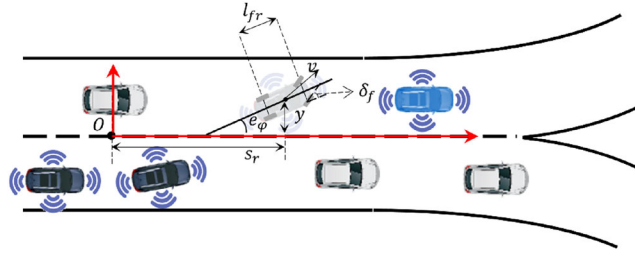
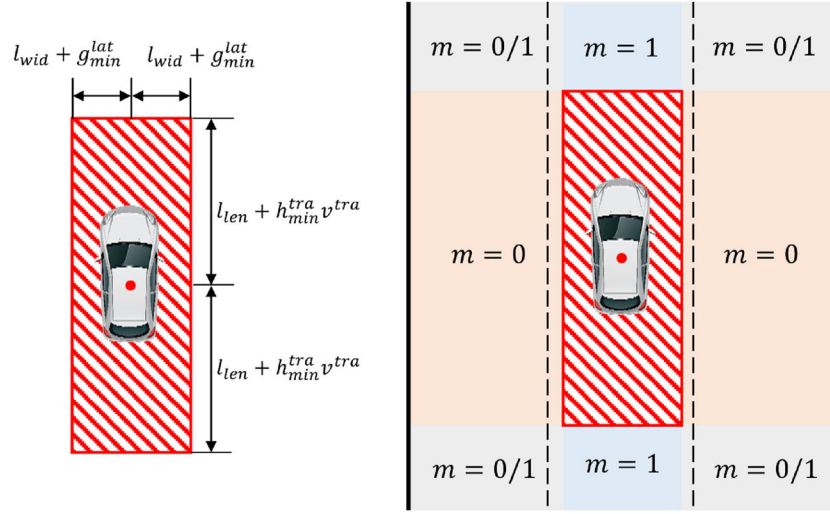


Figure 4. Scenario notations.

Figure 5. Value of m illustration.

$$\xi_{i+1} = A_i \xi_i + B_i u_i + C_i \quad (15)$$

$$A_i \triangleq \Delta S_R \times \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix} + I_{4 \times 4} \quad (16)$$

$$B_i \triangleq \Delta S_R \times \begin{bmatrix} -\pi_i^3 & 0 \\ 0 & 0 \\ 0 & 1 \\ 0 & \frac{1}{l_{fr}} \end{bmatrix} \quad (17)$$

$$C_i \triangleq \Delta S_R \times [0 \ 0 \ -\kappa \ 0]^T \quad (18)$$

Equations (15–17), and (18) describe the vehicle dynamics for the proposed controller.

Cost function

The objective of the proposed SuPLC controller is to minimize total state error in the control horizon. The state cost is formulated into a quadratic form $\frac{1}{2}(\xi_i - \xi_{desi})^T Q(\xi_i - \xi_{desi})$. The control cost $\frac{1}{2}u_i^T R u_i$ is used to restrict control efforts. The objective function is formulated as follows:

$$\begin{aligned} \min_u \left(\underbrace{\sum_{i=1}^N \left(\frac{1}{2}(\xi_i - \xi_{desi})^T Q(\xi_i - \xi_{desi}) \right)}_{\text{State error cost}} \right. \\ \left. + \underbrace{\frac{1}{2}u_i^T R u_i}_{\text{Control cost}} + \underbrace{\frac{1}{2}(\xi_{N+1} - \xi_{desN+1})^T Q(\xi_{N+1} - \xi_{desN+1})}_{\text{Terminal state error cost}} \right) \end{aligned} \quad (19)$$

where ξ_{desi} is the desired state at step i ; ξ_{N+1} is the terminal state. Weighting factors Q and R are defined as follows:

$$Q \triangleq \begin{bmatrix} q_1 & 0 & 0 & 0 \\ 0 & q_2 & 0 & 0 \\ 0 & 0 & q_3 & 0 \\ 0 & 0 & 0 & q_4 \end{bmatrix} \quad (20)$$

$$R \triangleq \begin{bmatrix} r_1 & 0 \\ 0 & r_2 \end{bmatrix} \quad (21)$$

where q_1, q_2, q_3, q_4, r_1 , and r_2 are non-negative constants.

Constraints

The proposed controller is bounded by collision avoidance, desired speed range, geometry boundaries, acceleration capability, and yaw angle constraints.

Collision avoidance constraints

Longitudinal collision avoidance between adjacent CAVs in a platoon is achieved by regulating time headway as follows:

$$t^j(s_r) - t^{j-1}(s_r) > h_{min}^p \quad (22)$$

where h_{min}^p is the safe following headway in a platoon.

Lateral collision avoidance between a CAV and a surrounding HV is ensured by equations (23) and (24).

$$|y^k - y^j| > l_{wid} + g_{min}^{lat} - mM \quad (23)$$

$$|s_r^k - s_r^j| > l_{len} + h_{min}^{tra} v^{tra} - (1 - m)M \quad (24)$$

where k is the index of surrounding vehicles; g_{min}^{lat} is the lateral safe gap; l_{wid} is the width of a vehicle; l_{len} is the length of a vehicle; v^{tra} is the speed of traffic flow; h_{min}^{tra} is the safe following headway in traffic. m is a decision variable. $m \in \{0, 1\}$. M is a constant and significantly large value. The value of m is directly associated with the relative position of ego vehicle and lag vehicle, as demonstrated in Figure 5. The value of m is determined by finding the best relative position that minimizes the cost (equation (19)). This concept follows the big M method (Ding et al., 2014). The optimization solver adopted iterates through both potential values of m and picks a value that produces the minimal cost.

Geometry boundaries constraint

Vehicles should travel within road geometry boundaries as follows:

$$y_{min} \leq y \leq y_{max} \quad (25)$$

Acceleration constraint

The vehicle's acceleration should be bounded by vehicle capability and comfort.

$$a_{min} \leq a \leq a_{max} \quad (26)$$

Front-wheel angle constraint

The vehicle's front-wheel angle should be within its steering range.

$$\delta_{fmin} \leq \delta_f \leq \delta_{fmax} \quad (27)$$

Solution algorithm

A solution algorithm previously developed by this research group is borrowed (Yu Zhang, 2020). It was originally developed for the longitudinal automation of CACC. Here it is enhanced for longitudinal and lateral coupled automation functions. It can accelerate computation speed greatly by pre-determining the optimal terminal state. In each iteration cycle, this algorithm involves a backward calculation of concomitant matrices and a forward calculation of the control vector and state vector. It can iterate several times to decrease the effects of inconstant coefficient matrices. Moreover, inspired by the conventional branch and bound method (Lawler & Wood, 1966), this algorithm is further enhanced here: i) restricting the objective function into a form of quadratic terms only; ii) considering constraints; iii) considering parameter-varying dynamics coefficients; iv) capable of solving mixed-integer programming problems. These designs are better compatible with the problem of interest. This algorithm is described in Table 2.

Performance analysis

String stability is a must for a CACC system to resist speed perturbations. It has been proposed and proven by past studies on longitudinal only CACC. However, the proposed SuPLC controller is longitudinal and lateral coupled. Followers are with a goal of tracking the lateral position trajectory of the leader. Therefore, lateral stability should also be proven to ensure lateral oscillation would not be amplified when propagating along a vehicle string. Moreover, the lane-change capacity of the proposed SuPLC controller is quantified theoretically in this section.

Pontryagin's minimum principle (PMP) is a method to solve optimal control problems (Khalil, 1994; Saerens et al., 2013; Wang et al., 2016). It is presented in Definition 3 for further analysis.

Definition 3 (PMP method): Necessary conditions for the optimal solution U^* are as follows:

$$H(X, U, \lambda, s_r) \stackrel{\text{def}}{=} L(X, U, s_r) + \lambda^T f(X, U, s_r) \quad (28)$$

$$H(X^*, U^*, \lambda^*, s_r) \leq H(X, U, \lambda, s_r) \quad (29)$$

$$\Psi(X, S) + H(X, U, \lambda, S) = 0 \quad (30)$$

$$-\frac{d\lambda}{ds_r} = \frac{\partial H}{\partial X} = \frac{\partial f}{\partial X} \lambda + \frac{\partial L}{\partial X} \quad (31)$$

$$\lambda(S) = \frac{\partial \Psi(X, S)}{\partial X(S)} \quad (32)$$

Table 2. Solution algorithm.**Input:** initial state ξ_0 , desired state ξ_{des} , $\mathbf{Q}$, $\mathbf{R}$, number of control steps N **Output:** optimal control $\mathbf{u}_i$ and state ξ_i for each step**Initialize:** $iter = 0$ **while** $\|\mathbf{u}^{iter} - \mathbf{u}^{iter-1}\| > \varepsilon$ **do** $iter = iter + 1$ use $\mathbf{u}^{iter-1}$ to calculate $\mathbf{A}_i$, $\mathbf{B}_i$, and $\mathbf{C}_i$ for $i \in \{0, 1, \dots, N\}$ initialize $J^* \leftarrow \infty$, the domain of $m_i : \Theta \in (-\infty, \infty)$ **while** Θ is not empty **do**pick and remove $m_i \in \{0, 1\}$ from Θ $\sim \mathbf{u}, \sim \mathbf{m}, J(\sim \mathbf{u}, \sim \mathbf{m}) \leftarrow$ solve QP problem $QP(\Theta)$:

$$\sim \mathbf{Q}_{N+1} \triangleq \mathbf{Q} \quad (132)$$

$$\sim \mathbf{D}_{N+1} \triangleq -\mathbf{Q}\xi_{des, N+1} \quad (133)$$

For $i \in \{N, N-1, \dots, 1\}$

$$\mathbf{D}_i = -\mathbf{Q}\xi_{des, i} \quad (134)$$

$$\sim \mathbf{Q}_i = \mathbf{Z}_{2i}^T \mathbf{R}_i \mathbf{Z}_{2i} + \mathbf{Z}_{4i}^T \sim \mathbf{Q}_{i+1} \mathbf{Z}_{4i} + \mathbf{Q}_i \quad (135)$$

$$\sim \mathbf{D}_i = \mathbf{Z}_{2i}^T \mathbf{R}_i \mathbf{Z}_{3i} + \mathbf{Z}_{4i}^T \sim \mathbf{Q}_{i+1} \mathbf{Z}_{5i} + \mathbf{Z}_{4i}^T \sim \mathbf{D}_{i+1} + \mathbf{D}_i \quad (136)$$

$$\mathbf{Z}_{1i} = (\mathbf{R}_i + \mathbf{B}_i^T \sim \mathbf{Q}_{i+1} \mathbf{B}_i)^{-1} \quad (137)$$

$$\mathbf{Z}_{2i} = -\mathbf{Z}_{1i} \mathbf{B}_i^T \sim \mathbf{Q}_{i+1} \mathbf{A}_i \quad (138)$$

$$\mathbf{Z}_{3i} = -\mathbf{Z}_{1i} \mathbf{B}_i^T (\sim \mathbf{Q}_{i+1} \mathbf{C}_i + \sim \mathbf{D}_{i+1}) \quad (139)$$

$$\mathbf{Z}_{4i} = \mathbf{A}_i + \mathbf{B}_i \mathbf{Z}_{2i} \quad (140)$$

$$\mathbf{Z}_{5i} = \mathbf{B}_i \mathbf{Z}_{3i} + \mathbf{C}_i \quad (141)$$

For $i \in \{0, 1, \dots, N\}$

$$\mathbf{u}_i = \mathbf{Z}_{2i} \xi_i + \mathbf{Z}_{3i} \quad (142)$$

$$\xi_{i+1} = \mathbf{Z}_{4i} \xi_i + \mathbf{Z}_{5i} \quad (143)$$

if $QP(\Theta)$ is infeasible **then**

prune current node

else if $J(\sim \mathbf{u}, \sim \mathbf{m}) > J^*$ **then**

prune current node

else if $\sim \mathbf{m} \in \{0, 1\}$ **then** $J^* \leftarrow J(\sim \mathbf{u}, \sim \mathbf{m}), \mathbf{u}^{iter} \leftarrow \sim \mathbf{u}$ **Else**choose integer $\hat{m}_i : m_i \in \{0, 1\}$ from $\sim \mathbf{m}$ **if** $\hat{m}_i$ is feasible and $J(\hat{\mathbf{u}}, \hat{\mathbf{m}}) < J^*$ **then** $J^* \leftarrow J(\hat{\mathbf{u}}, \hat{\mathbf{m}}), \mathbf{u}^{iter} \leftarrow \hat{\mathbf{u}}$

branch node

if $\xi_{iter} > \xi_{max}$ **then** $\xi_{iter} = \xi_{max}$ **if** $\xi_{iter} < \xi_{min}$ **then** $\xi_{iter} = \xi_{min}$ **if** $\mathbf{u}^{iter} > \mathbf{u}_{max}$ **then** $\mathbf{u}^{iter} = \mathbf{u}_{max}$ **if** $\mathbf{u}^{iter} < \mathbf{u}_{min}$ **then** $\mathbf{u}^{iter} = \mathbf{u}_{min}$

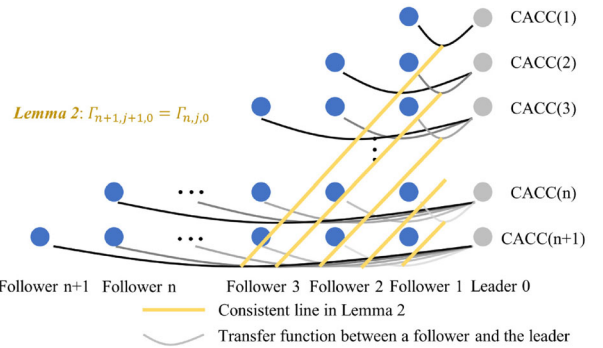
where $\mathbf{X}$ is state vector; $\mathbf{U}$ is control vector; λ is a co-state vector; L is the running cost function; f is the system dynamics function; $\mathcal{S}$ is control horizon; Ψ is the terminal cost function.

To analyze platoon performance, Proposition 1 is first adopted.

Proposition 1 (Platoon controller in continuous form): A centralized platoon controller is presented in the following. This platoon controller is equivalent to the SuPLC controller.

$$\min_{\mathbf{U}} J = \underbrace{\Psi(\mathbf{X}_n)}_{\text{Terminal cost}} + \underbrace{\int_0^{S_r} L(\mathbf{X}_n, \mathbf{U}_n) ds_r}_{\text{Running cost}} \quad (33)$$

$$\text{s.t. } f(\mathbf{X}_n, \mathbf{U}_n) \triangleq \frac{\partial \mathbf{X}_n(S_r)}{\partial S_r} = \mathbb{A}_n \mathbf{X}_n(S_r) + \mathbb{B}_n \mathbf{U}_n(S_r) + \mathbb{C}_n \quad (34)$$

**Figure 6.** Relation of the transfer function between a follower and the leader.

where

$$L(\mathbf{X}_n, \mathbf{U}_n) \triangleq \frac{1}{2} \mathbf{X}_n^T(S_r) \mathbf{Q}_n \mathbf{X}_n(S_r) + \frac{1}{2} \mathbf{U}_n^T(S_r) \mathbf{R}_n \mathbf{U}_n(S_r) \quad (35)$$

$$\mathbf{X}_n(s_r) \stackrel{\text{def}}{=} \begin{bmatrix} \xi^1(s_r) - \xi_{des}^1(s_r) \\ \xi^2(s_r) - \xi_{des}^2(s_r) \\ \vdots \\ \xi^n(s_r) - \xi_{des}^n(s_r) \end{bmatrix}_{4n \times 1} \quad (36)$$

$$\mathbf{U}_n(s_r) \stackrel{\text{def}}{=} \begin{bmatrix} \mathbf{u}^1(s_r) \\ \mathbf{u}^2(s_r) \\ \vdots \\ \mathbf{u}^n(s_r) \end{bmatrix}_{2n \times 1} \quad (37)$$

$$\mathbf{Q}_n \stackrel{\text{def}}{=} \begin{bmatrix} \mathbf{Q} & 0 & \cdots & 0 \\ 0 & \mathbf{Q} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \mathbf{Q} \end{bmatrix}_{4n \times 4n} \quad (38)$$

$$\mathbf{R}_n \stackrel{\text{def}}{=} \begin{bmatrix} \mathbf{R} & 0 & \cdots & 0 \\ 0 & \mathbf{R} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \mathbf{R} \end{bmatrix}_{2n \times 2n} \quad (39)$$

$$\mathbf{A}_n \stackrel{\text{def}}{=} \begin{bmatrix} \mathbf{A} & 0 & \cdots & 0 \\ 0 & \mathbf{A} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \mathbf{A} \end{bmatrix}_{4n \times 4n} \quad (40)$$

$$\mathbf{B}_n \stackrel{\text{def}}{=} \begin{bmatrix} \mathbf{B} & 0 & \cdots & 0 \\ 0 & \mathbf{B} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \mathbf{B} \end{bmatrix}_{4n \times 2n} \quad (41)$$

$$\mathbf{A} \stackrel{\text{def}}{=} \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix} \quad (42)$$

$$\mathbf{B} \stackrel{\text{def}}{=} \begin{bmatrix} -(\pi^0(0))^3 & 0 \\ 0 & 0 \\ 0 & 1 \\ 0 & \frac{1}{l_{fr}} \end{bmatrix} \quad (43)$$

$$\mathbf{C}_n \stackrel{\text{def}}{=} \mathbf{C}_n^{Str} \mathbf{a}^0 - \mathbf{C}_n^{Lat} \delta_f^0 \quad (44)$$

$$\mathbf{C}_n^{Str} \stackrel{\text{def}}{=} \underbrace{\begin{bmatrix} (\pi^0(0))^3 & 0 & 0 & 0 \end{bmatrix}^T}_n \quad (45)$$

$$\mathbf{C}_n^{Lat} \stackrel{\text{def}}{=} \underbrace{\begin{bmatrix} 0 & 0 & \frac{1}{l_{fr}} & 0 \end{bmatrix}^T}_n \quad (46)$$

Stability criteria

String stability and lateral stability describe the condition that state oscillation would not be amplified backward in a platoon. Since oscillation is better analyzed in the frequency domain, the Laplace transform is adopted as presented in Definition 4. Stability criteria are then proposed in Definition 5.

Definition 4: (Laplace transform): Function $g(x)$ with $x \geq 0$ can be transformed into the frequency domain as follows:

$$G(s) = \mathcal{L}\{g(x)\} = \int_0^\infty g(x)e^{-sx}dx \quad (47)$$

$$s \stackrel{\text{def}}{=} i\omega (\omega \in \mathbb{R}^+) \quad (48)$$

where e denotes the natural logarithm; s is a complex variable; i denotes imaginary number; ω denotes angular speed in the Laplace domain.

By adopting existing string stability definitions (Ploeg et al., 2014), string stability and lateral stability criteria are defined as follows:

Definition 5 (Stability criteria): For $\forall n \in \mathbb{N}^+$, the platoon is string stable and lateral stable if and only if:

$$\Gamma_{n,j,j-1}^{Str} \stackrel{\text{def}}{=} \frac{\|a^j(s)\|_2}{\|a^{j-1}(s)\|_2} \leq 1, \quad \forall j \in [1, n], \quad \forall n \in \mathbb{N}^+ \quad (49)$$

$$\Gamma_{n,j,j-1}^{Lat} \stackrel{\text{def}}{=} \frac{\|\delta_f^j(s)\|_2}{\|\delta_f^{j-1}(s)\|_2} \leq 1, \quad \forall j \in [1, n], \quad \forall n \in \mathbb{N}^+ \quad (50)$$

where $\Gamma_{n,j,j-1}^{Str}$ is the acceleration oscillation transfer function between follower j and follower $j-1$; $\Gamma_{n,j,j-1}^{Lat}$ is the front wheel angle oscillation transfer function between follower j and follower $j-1$; $\|\cdot\|_2$ denotes 2-norm.

String stability and lateral stability analysis

Lemmas 1–4 are first proposed. String stability and lateral stability are then proven in Theorem 1. The logic of analysis is presented as follows:

Lemma 1 presents the oscillation transfer function $\Gamma_{n,j,0}$ between the follower j and the leader.

Lemma 2 discusses the propagation rule ($\Gamma_{n,j,0}$, transfer function between follower j and the leader) with regard to various platoon sizes, as illustrated in the yellow line in Figure 6.

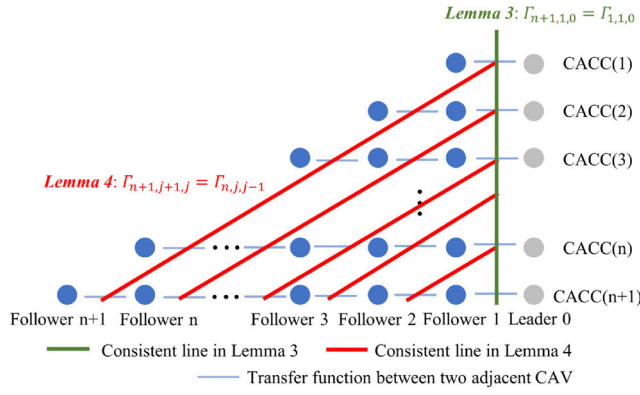


Figure 7. Relation of the transfer function between adjacent CAVs.

Lemma 3 discusses the propagation rule ($\Gamma_{n,1,0}$, transfer function between the first follower and the leader) with regard to various platoon sizes, as illustrated in the green line in **Figure 7**.

Lemma 4 discusses the propagation rule ($\Gamma_{n,j,j-1}$, transfer function between two adjacent followers) with regard to various platoon sizes, as illustrated in the red line in **Figure 7**.

Theorem 1 proves that the oscillation transfer function is always less than 1 ($\Gamma_{n,j,j-1} = \Gamma_{1,1,0} < 1$). It concludes the stability analysis.

Lemma 1 (Follower-leader transfer function ($\Gamma_{n,j,0}$)): Oscillation transfer functions between follower j and the platoon leader are as follows:

$$\Gamma_{n,j,0}^{Str} = \|\mathbf{O}_{n,j}^{Str}(\mathbf{R}_n - \mathbf{M}_n \mathbb{B}_n)^{-1} \mathbf{M}_n \mathbf{C}_n^{Str}\|_2 \quad (51)$$

$$\Gamma_{n,j,0}^{Lat} = \|\mathbf{O}_{n,j}^{Lat}(\mathbf{R}_n - \mathbf{M}_n \mathbb{B}_n)^{-1} \mathbf{M}_n \mathbf{C}_n^{Lat}\|_2 \quad (52)$$

where

$$\mathbf{O}_{n,j}^{Str} \triangleq \begin{bmatrix} 0 \cdots 0 \\ \underbrace{\quad}_{2j-2} \quad 1, 0 \quad 0 \cdots 0 \\ \underbrace{\quad}_{2n-2j} \quad \underbrace{\quad}_{1 \times 2n} \end{bmatrix} \quad (53)$$

$$\mathbf{O}_{n,j}^{Lat} \triangleq \begin{bmatrix} 0 \cdots 0 \\ \underbrace{\quad}_{2j-2} \quad 0, 1 \quad 0 \cdots 0 \\ \underbrace{\quad}_{2n-2j} \quad \underbrace{\quad}_{1 \times 2n} \end{bmatrix} \quad (54)$$

$$\mathbf{E}_n \triangleq (\mathbf{sI} + \mathbb{A}_n^T)^{-1} \quad (55)$$

$$\mathbf{F}_n \triangleq (\mathbf{sI} - \mathbb{A}_n)^{-1} \quad (56)$$

$$\mathbf{M}_n \triangleq \mathbb{B}_n^T \mathbf{E}_n \mathbf{Q}_n \mathbf{F}_n \quad (57)$$

Lemma 2 (The propagation rule ($\Gamma_{n,j,0}$)): With regard to various platoon size n , the oscillation transfer

function $\Gamma_{n,j,0}$ between follower j and the leader follows the following propagation rule:

$$\Gamma_{n+1,j+1,0}^{Str} \equiv \Gamma_{n,j,0}^{Str} \quad (58)$$

$$\Gamma_{n+1,j+1,0}^{Lat} \equiv \Gamma_{n,j,0}^{Lat} \quad (59)$$

Lemma 3 (The propagation rule ($\Gamma_{n,1,0}$)): The oscillation transfer function $\Gamma_{n,1,0}$ between the first follower and the leader is constant over various platoon size n :

$$\Gamma_{n+1,1,0}^{Str} \equiv \Gamma_{1,1,0}^{Str} \quad (60)$$

$$\Gamma_{n+1,1,0}^{Lat} \equiv \Gamma_{1,1,0}^{Lat} \quad (61)$$

Lemma 4 (The propagation rule ($\Gamma_{n,j,j-1}$)): With different platoon size n , the propagation rule of the transfer function $\Gamma_{n,j,j-1}$ between two adjacent followers is as follows:

$$\Gamma_{n+1,j+1,j}^{Str} \equiv \Gamma_{n,j,j-1}^{Str} \quad (62)$$

$$\Gamma_{n+1,j+1,j}^{Lat} \equiv \Gamma_{n,j,j-1}^{Lat} \quad (63)$$

Theorem 1 (String stability and lateral stability): The CACC platoon under control of the proposed SuPLC controller is with string stability and lateral stability.

Proof: As illustrated in **Figure 7**, **Lemmas 3** and **4** give:

$$\Gamma_{n,j,j-1}^{Str} = \Gamma_{1,1,0}^{Str} \quad (64)$$

$$\Gamma_{n,j,j-1}^{Lat} = \Gamma_{1,1,0}^{Lat} \quad (65)$$

Since q_1, q_2, q_3, q_4, r_1 , and r_2 are all non-negative, the following inequalities are always satisfied.

$$\Gamma_{1,1,0}^{Str} = \left\| \frac{2q_1(\pi^0(0))^6}{2q_1(\pi^0(0))^6 + r_1} \right\|_2 \leq 1 \quad (66)$$

$$\Gamma_{1,1,0}^{Lat} = \left\| \frac{2q_3}{2q_3 + r_2 l_{fr}^2} \right\|_2 \leq 1 \quad (67)$$

Applying **equations (66)** and **(67)** into **equations (64)** and **(65)**:

$$\Gamma_{n,j,j-1}^{Str} \leq 1 \quad (68)$$

$$\Gamma_{n,j,j-1}^{Lat} \leq 1 \quad (69)$$

The stability criteria in **Definition 5** are satisfied. Therefore, the CACC platoon under control of the proposed SuPLC controller is string stable and lateral stable regardless of platoon size. This concludes the proof. $\square$

Platoon lane-change capacity

Compared with the conventional SiPLC controller, the proposed SuPLC controller is capable of maneuvering a platoon through smaller windows. Therefore, a moving lane-change window that is large enough for a single CAV's lane-change maneuver is also theoretically sufficient for platoon lane-change maneuver. The minimum lane-change window reduction enhances the maximum platoon lane-change capacity. In this section, this enhancement on lane-change capacity is quantified.

Platoon lane-change capacity P_{PLC} is defined as the probability of traffic headway larger than the minimum platoon lane-change headway h_{min}^{PLC} :

$$P_{PLC} = P(h \geq h_{min}^{PLC}) \quad (70)$$

A traffic headway distribution model is utilized to compute this probability as follows:

$$P(h \geq h_{min}^{PLC}) = \exp\left(-\frac{h_{min}^{PLC} - h_{min}^{tra}}{\bar{h}^{tra} - h_{min}^{tra}}\right) \quad (71)$$

where h_{min}^{tra} is the minimum traffic headway; $\bar{h}^{tra}$ is average traffic time headway:

$$\bar{h}^{tra} = \frac{3600}{V} \quad (72)$$

where V is traffic volume in the unit of $veh/hr/ln$. It is computed as follows (Margiotto & Washburn, 2017):

$$V = v/c \left(2200 + 10 \times (\min(70, v_{free}^{tra}) - 50) \right) \quad (73)$$

where v/c is volume-to-capacity ratio; v_{free}^{tra} is the free-flow speed in the unit of mph .

Therefore, the platoon lane-change capacity is finally quantified as follows:

$$P_{PLC} = \exp\left(-\frac{(h_{min}^{PLC} - h_{min}^{tra})v/c(2200 + 10 \times (\min(70, v_{free}^{tra}) - 50))}{3600 - h_{min}^{tra}v/c(2200 + 10 \times (\min(70, v_{free}^{tra}) - 50))}\right) \quad (74)$$

The minimum successive platoon lane-change headway h_{min}^{SuPLC} and the minimum simultaneous platoon lane-change headway h_{min}^{SiPLC} are computed as follows:

$$h_{min}^{SuPLC} = h_{min}^{LC} \quad (75)$$

$$h_{min}^{SiPLC} = nh_{des}^p + h_{min}^{LC} \quad (76)$$

where h_{min}^{LC} is the safe lane-change headway of a single vehicle; h_{des}^p is the desired headway in a platoon.

Following settings are adopted:

- h_{min}^{tra} is set to 1.1 sec (Milanés & Shladover, 2014);
- h_{des}^p is set to 0.5 sec (Milanés & Shladover, 2014);

- h_{min}^{LC} is set to 2.68 sec (Yang et al., 2019).

The platoon lane-change capacity of the proposed SuPLC controller P_{SuPLC} and the platoon lane-change capacity of conventional SiPLC controller P_{SiPLC} are computed as follows:

$$P_{SuPLC} = \exp\left(-\frac{1.58v/c(2200 + 10 \times (\min(70, v_{free}^{tra}) - 50))}{3600 - 1.1v/c(2200 + 10 \times (\min(70, v_{free}^{tra}) - 50))}\right) \quad (77)$$

$$P_{SiPLC} = \exp\left(-\frac{(0.5n + 1.58)v/c(2200 + 10 \times (\min(70, v_{free}^{tra}) - 50))}{3600 - 1.1v/c(2200 + 10 \times (\min(70, v_{free}^{tra}) - 50))}\right) \quad (78)$$

A sensitivity analysis is conducted by numerical simulation. It considers two scenarios: i) arterial with a free-flow speed of 72 km/h (45 mph); ii) freeway with a free-flow speed of 105 km/h (65 mph).

Simulation results are illustrated in Figure 8. It demonstrates that the SuPLC controller enhances platoon lane-change capacity by 20.44% on arterials and 18.81% on freeways, compared to the SiPLC controller.

Evaluations

The proposed SuPLC controller is evaluated from the following aspects: i) platoon lane-change capability; ii) successive platoon lane-change function validation; iii) platoon string stability; iv) platoon lateral stability. Following settings are adopted for the evaluation.

- Following headway in a platoon h_{min}^p : 0.7s;
- Acceleration range: $[-5, 3]m/s^2$;
- Steering wheel angle range: $[-450^\circ, 450^\circ]$;
- Lane width: 3m (9.8ft);
- Human reaction time t_{react}^{HV} : 1.5s;
- CAV reaction time t_{react}^{CAV} : 0.5s;
- Delay of brake $t_{brake, delay}$: 0.35s.

Function verification with vehicle dynamics

A simulation is conducted on a joint simulation platform consisting of PreScan and Matlab/Simulink. PreScan follows a systematic physics-based simulation approach. Efficient parametric math models reproduce system-level vehicle dynamics behavior. Therefore, this experiment is a function verification with vehicle dynamics embedded into the platform. The goal of this evaluation is as follows:

- **Function validation:** This test is to verify the function of changing lane successively from the aspect of vehicle control performance.

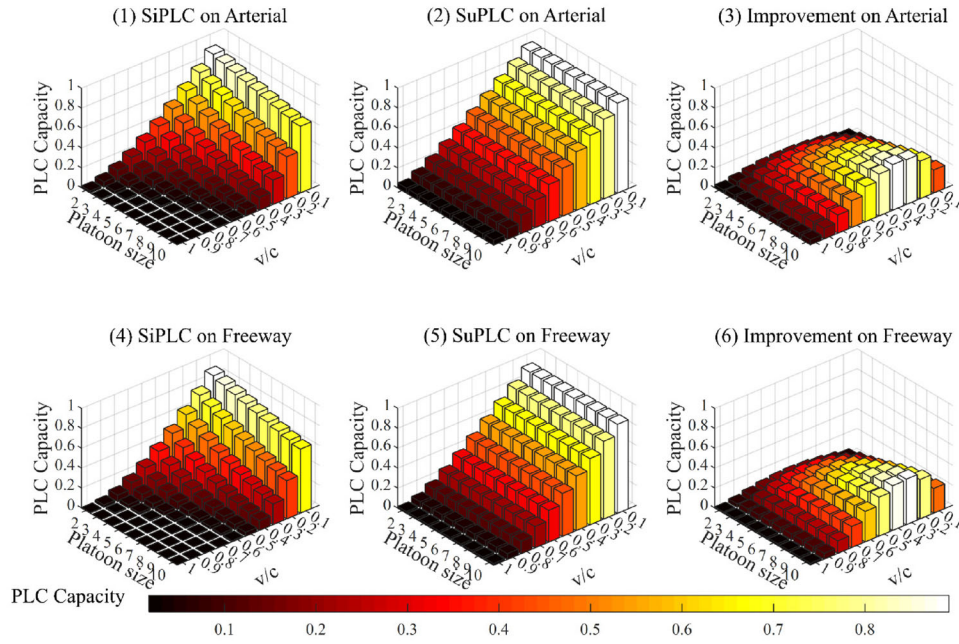


Figure 8. Platoon lane-change capacity analysis.

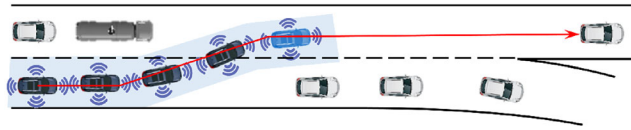


Figure 9. Scenario 1: successive platoon lane-change function validation.

- **Stability evaluation:** This test is to evaluate the string stability and lateral stability of the proposed controller.

Test scenarios

The test is designed on two road types:

- **Arterial:** with a speed limit of 45mph (72km/h);
- **Freeway:** with a speed limit of 65mph (105km/h).

Two scenarios are designed for different evaluation purposes.

Scenario 1: successive platoon lane-change function validation:

As illustrated in Figure 9, in this scenario, a platoon has to change lane to avoid running into the off-ramp. The leading and left-side HVs cannot provide the platoon with big enough cut-through windows. Therefore, the platoon has to change lane successively under the control of the proposed SuPLC controller.

Scenario 2: stability evaluation:

As illustrated in Figure 10, in this scenario, a platoon is free-cruising on a single-lane road. $\pm 7.2\text{km/h}$ speed perturbation and $\pm 1\text{m}$ lateral perturbation are

applied on the platoon leader respectively, as illustrated in Figure 11.

Results

Results confirm that the proposed controller is: i) capable of maneuvering a platoon through traffic by utilizing a smaller lane-change window; ii) with string stability; iii) with lateral stability; iv) with 15 milliseconds average computation time on a laptop equipped with an Intel i7-8750H CPU.

Figures 12 and 13 verify that the proposed controller functions as promised. It is able to maneuver a platoon through traffic. Figure 12 illustrates the trajectory of a platoon lane-change process. Performance of the proposed controller on both arterial and freeway are demonstrated. As shown in the figures, all CAVs are able to change lane successfully while keeping a safe following gap and maintaining the form of a platoon. Figure 13 illustrates gap trajectories between CAVs and background vehicles (front vehicle and left lag vehicle). No collision occurs in the platoon lane-change process.

The string stability of the proposed controller is confirmed. Figure 14 illustrates the acceleration trajectory of the platoon. Figure 15 illustrates the speed

trajectory of the platoon. The two figures demonstrate that acceleration and speed trajectories of followers have lower peaks compared to the leader. Therefore, the proposed SuPLC controller is string stable.

The lateral stability of the proposed controller is also confirmed. Figure 16 illustrates the steering angle trajectory of the platoon. Figure 17 illustrates the lateral position trajectory of the platoon. The two figures demonstrate that the steering angle and position of followers have lower peaks compared to the leader. Therefore, the platoon under control of the proposed SuPLC controller is laterally stable.

Function verification with traffic

In order to verify the proposed SuPLC controller in traffic, an evaluation has been conducted using a simulation platform. The platform is able to simulate CACC platoon in the context of traffic and was published in *Transportation Research Part C: Emerging Technologies* (Cui et al., 2018; Lai et al., 2020). This evaluation compares the lane-change capability between the proposed SuPLC controller and conventional SiPLC controller (Hsu & Liu, 2008). The goal of the evaluation is detailed as follows:

- **Function validation:** This test is to verify the function of changing lane successively for the proposed controller.
- **Platoon lane-change capability evaluation:** This test is to quantify the advantage of the proposed SuPLC controller over the conventional SiPLC controller.



Figure 10. Scenario 2: stability evaluation.

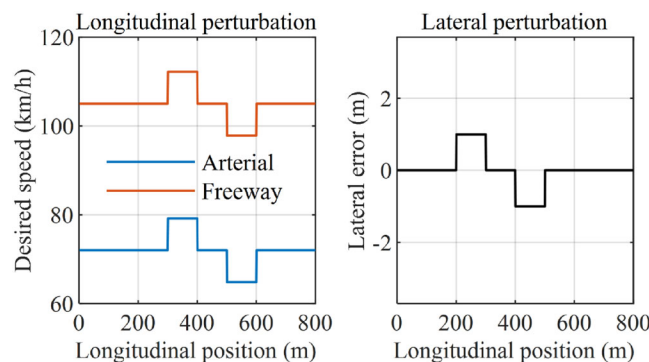


Figure 11. Longitudinal and lateral perturbation on the leader.

Test scenario

The test scenario is illustrated in Figure 18. In this scenario, the platoon is cruising in traffic. The platoon is with the objective of keeping its desired speed. Therefore, platoon lane-change decision is intrigued when the platoon is impeded by slow-moving HVs.

Eight cases are designed as shown in Table 3. Road types considered include arterial and freeway. Sensitivity analysis is conducted on congestion level (v/c ratio). For each case, 30 times simulations are conducted with different random seeds. The duration of each simulation run is 12 minutes.

Controller types

All surrounding HVs are controlled by driver models developed explicitly for human drivers (Cui et al., 2018; Lai et al., 2020). The platoon lane-change maneuver is controlled by two types of PLC controllers.

- **Baseline controller:** This controller is the conventional SiPLC controller. It maneuvers a platoon to change lane simultaneously.
- **The proposed controller:** This controller is the proposed SuPLC controller. It maneuvers a platoon to change lane successively. A conservative lane-change gap selection algorithm is adopted as shown in Table 1.

Measurements of effectiveness

The following measurements of effectiveness (MOEs) are adopted.

- **PLC competence:** This MOE is quantified by the average number of PLC per minute.
- **PLC efficiency:** This MOE is quantified by PLC delay. It is defined as the delay from the time when a PLC decision is made to the time when the PLC maneuver is finished. This delay includes two durations: LC window approaching and PLC maneuvering. The LC window approaching refers

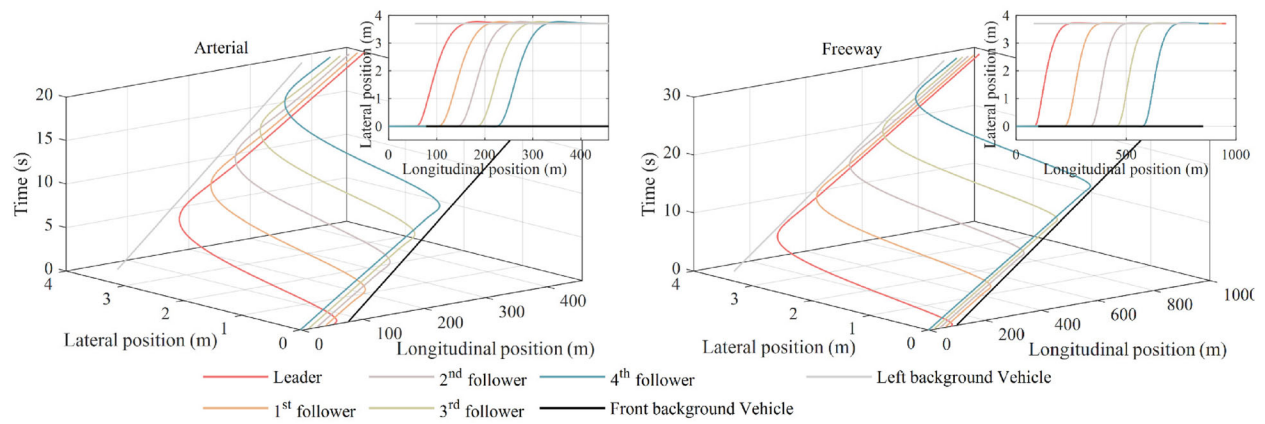


Figure 12. Vehicle trajectories.

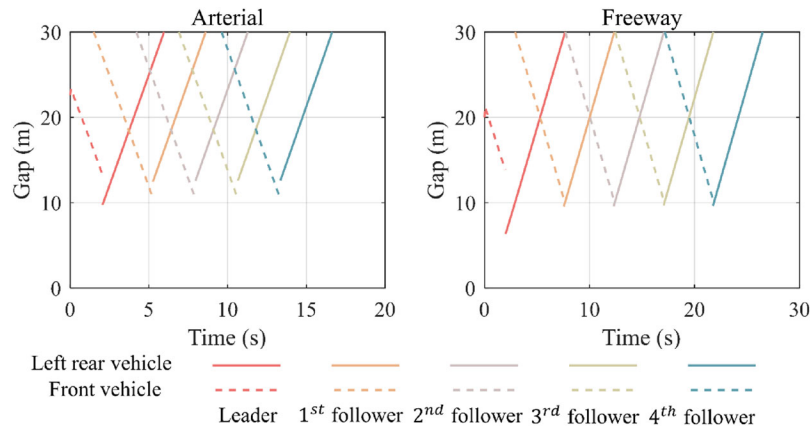


Figure 13. Gaps between CAVs and background vehicles.

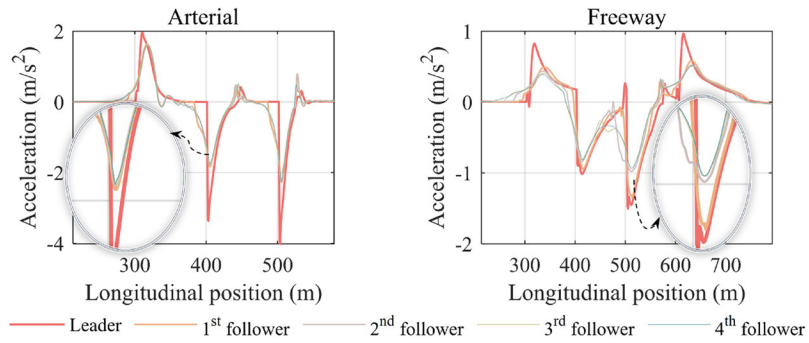


Figure 14. Acceleration trajectory.

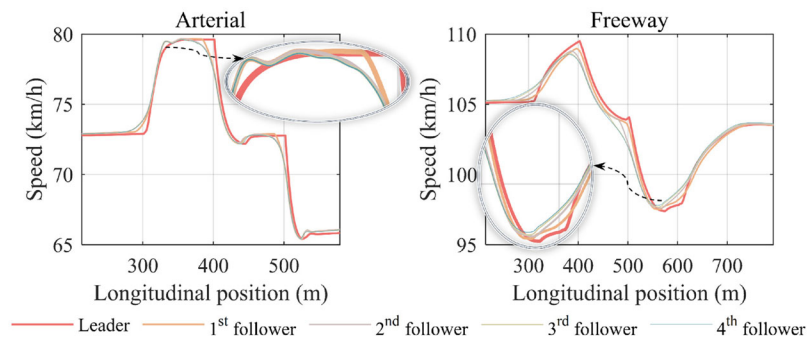


Figure 15. Speed trajectory.

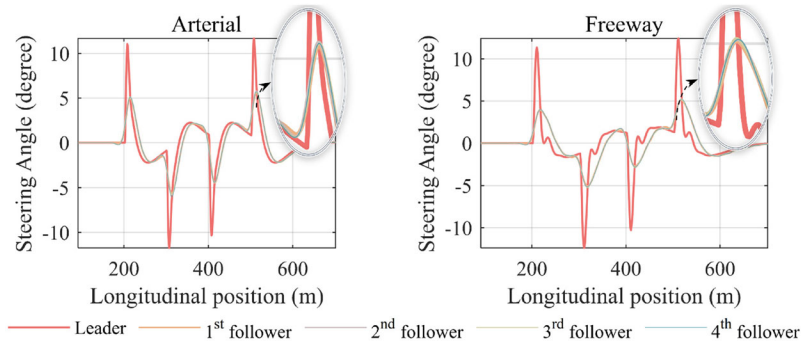


Figure 16. Steering angle trajectory.

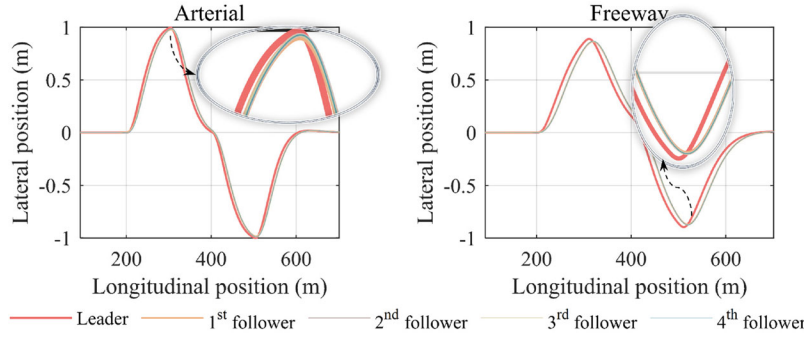


Figure 17. Lateral position trajectory.



Figure 18. Test scenario for PLC evaluation in traffic.

Table 3. Cases definition.

	$v/c = 0.25$	$v/c = 0.50$	$v/c = 0.75$	$v/c = 1.0$
Arterial $v_{\max} = 45$ mph(72 km/h), $v^{\text{tra}} = 37$ mph(60 km/h), $v_{\text{des}}^p = 45$ mph(72 km/h)	Case1	Case2	Case3	Case4
Freeway $v_{\max} = 65$ mph(105 km/h), $v^{\text{tra}} = 56$ mph(90 km/h), $v_{\text{des}}^p = 65$ mph(105 km/h)	Case5	Case6	Case7	Case8

to the process of a platoon approaching a target window in order to make a PLC maneuver.

Results

Results confirm that the proposed controller is: i) with successive platoon lane-change function; ii) superior to the conventional SiPLC controller.

Figures 19 and 20 illustrate trajectories from Case 5 under the control of SuPLC and SiPLC respectively. Under the control of the proposed controller, the platoon fulfills successive platoon lane-change maneuvers as demonstrated in Figure 19. Under the control of the conventional SiPLC controller, the platoon changes lane simultaneously as demonstrated in Figure 20. Moreover, over 95% SuPLC maneuvers are not impeded by traffic including cases with high congestion levels. It indicates that the proposed controller

could be adopted in field implementation. This result is reasonable since human drivers may not feel stressed and impeded when a faster vehicle cuts in. The SuPLC maneuver may not reduce the mobility of lag vehicles.

Figure 21 illustrates the average number of platoon lane changes per minute for all cases. It confirms that the proposed SuPLC controller does enhance the PLC competence no matter the road type under all v/c ratios. The proposed SuPLC controller enhances PLC competence by 120.49% on freeways and 71.36% on arterials. It means that the performance of the proposed SuPLC controller is positively correlated with speed. It makes sense as the proposed SuPLC controller is better at taking advantage of the marginal increase of distance headway due to speed change. The increase in traffic speed leads to a decrease in

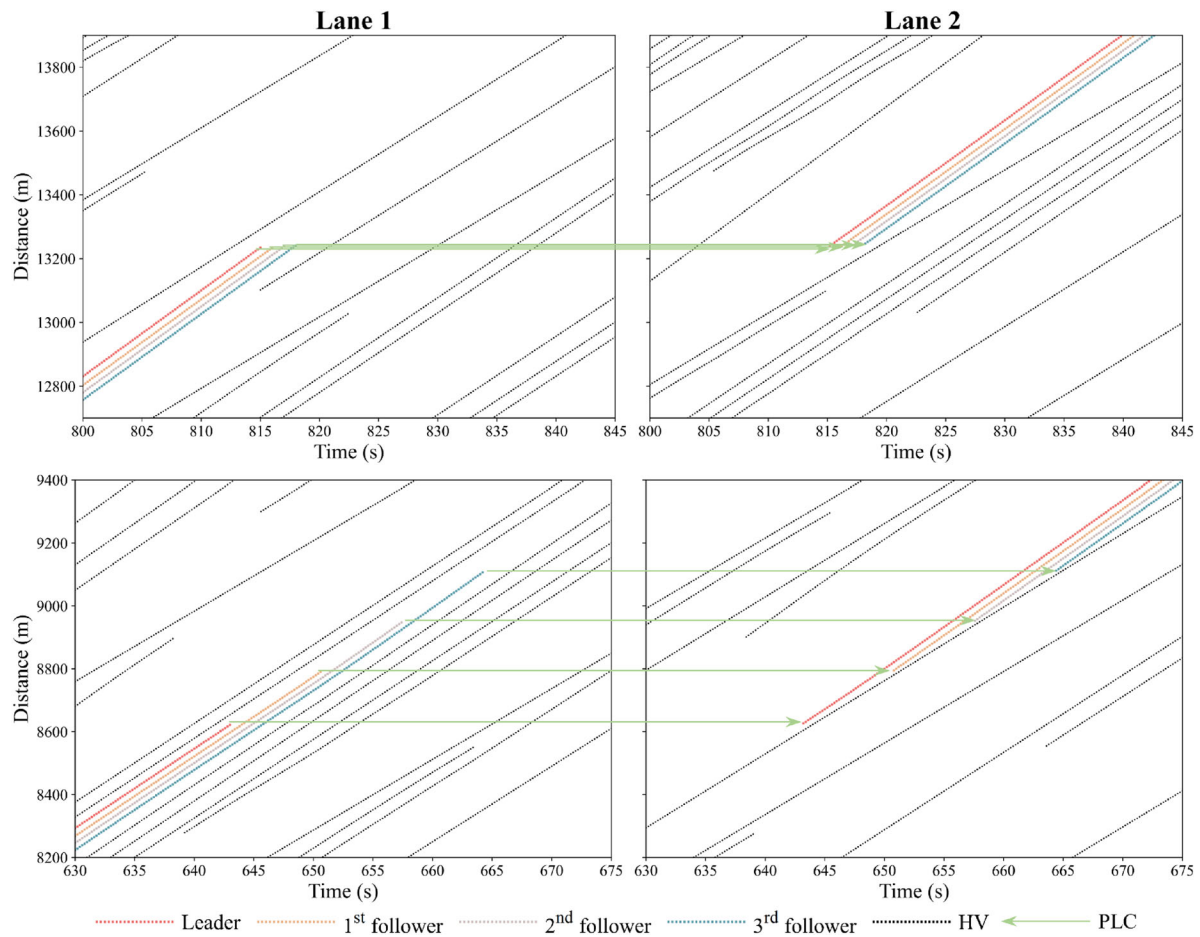


Figure 19. A sample trajectory of SuPLC.

density. The decrease of density may provide a larger distance window for PLC maneuver. Since the proposed SuPLC controller needs a much smaller PLC window, a slight increase in headway may be enough for the SuPLC maneuver, while insufficient for the SiPLC maneuver.

Figure 22 illustrates platoon lane-change delay distributions for all cases. It confirms that the proposed SuPLC controller does enhance PLC efficiency, compared to the conventional SiPLC controller. The proposed SuPLC controller reduces PLC delay by 41.36% on freeways and 25.05% on arterials. Therefore, the proposed SuPLC controller has a higher PLC efficiency enhancement on freeways than arterials. In addition, more outliers are found with the SuPLC controller. It makes sense as the performance of the proposed SuPLC controller heavily relies on the driving behavior of surrounding HVs. In the case that the surrounding vehicles form a constant barrier preventing the platoon from cutting through, there is just nothing that could be done even for the proposed SuPLC controller. Since the efficiency of the SuPLC controller is generally superior, more outliers are

observed. It indicates that the path planner of the leading vehicle is critical to PLC efficiency. Future studies could consider enhancing the path planner to avoid being trapped in a constant vehicle barrier.

A sensitivity analysis of performance enhancement to v/c ratio is presented in Figure 23. Performances have been demonstrated in regard to both PLC competence and PLC efficiency. The results show that the performance of the proposed SuPLC controller deteriorates with the increase of v/c ratio. However, the proposed SuPLC controller is always superior to the conventional SiPLC controller under all v/c ratios. It means the proposed SuPLC function could always be turned on in real implementation. In addition, the performance enhancement of the proposed SuPLC controller peaks when v/c ratio is between 0.5 and 0.75. This indicates that the proposed controller is more robust against congestion.

Conclusion and future research

This research proposes a CACC controller with platoon lane-change capability. The proposed SuPLC

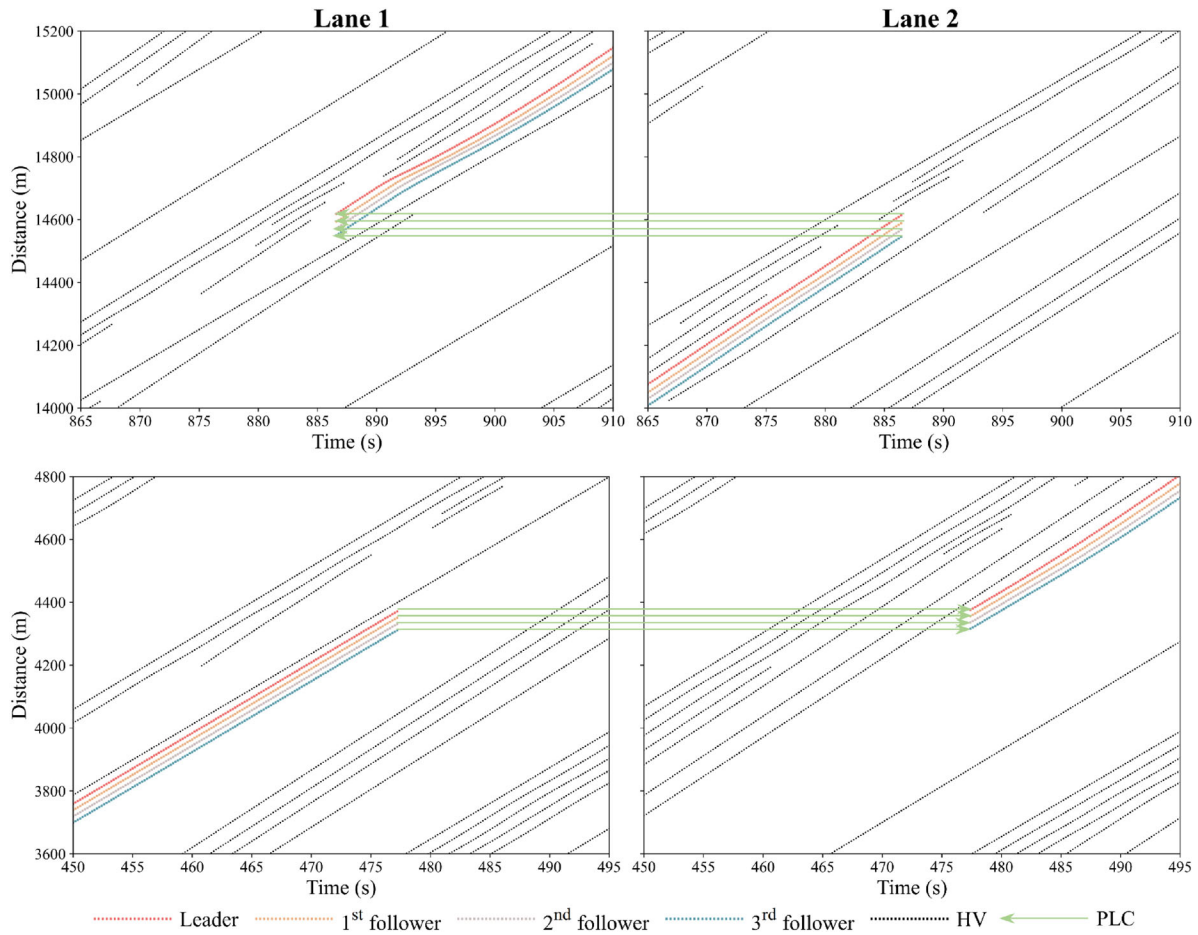


Figure 20. A sample trajectory of SiPLC.

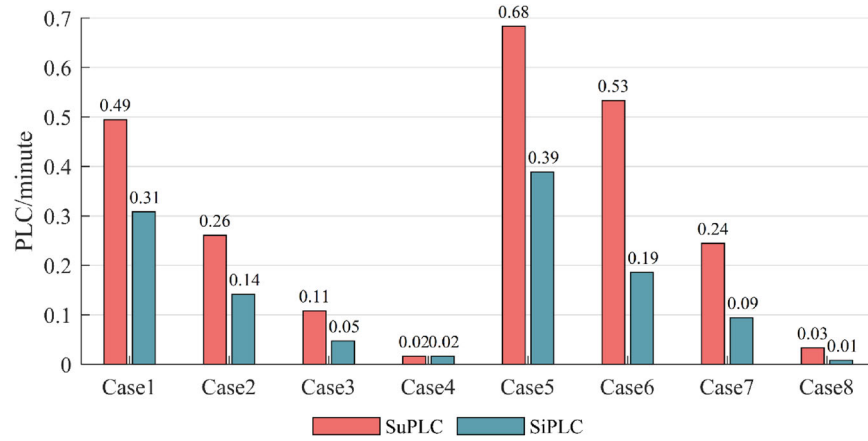


Figure 21. Platoon lane-change competence.

controller is designed for maneuvering a platoon to change lane successively like a snake. By adopting smaller lane-change gaps, the proposed controller improves the lane-change capability of CACC. The proposed controller has the following features: i) with successive platoon lane-change capability; ii) with string stability and lateral stability; iii) with consideration of vehicle dynamics. The proposed controller is

evaluated on a simulation platform with the context of traffic and a joint simulation platform consisting of PreScan and Matlab/Simulink. Results demonstrate that:

- The proposed SuPLC controller functions as expected and is able to maneuver a platoon through traffic via small windows.

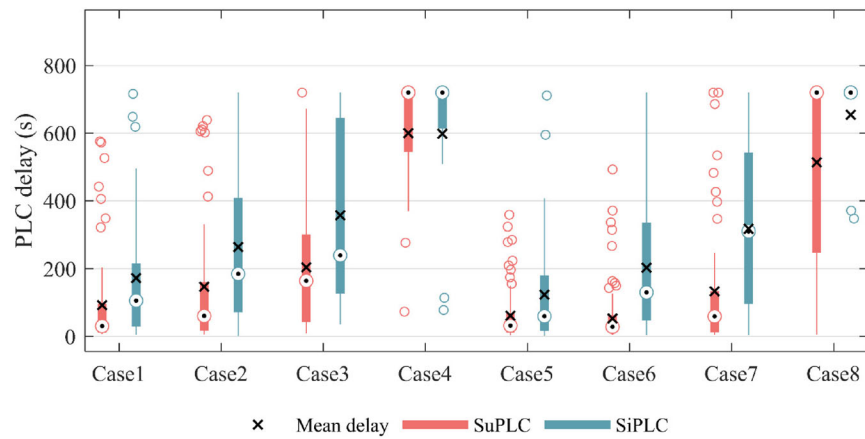


Figure 22. Platoon lane-change efficiency.

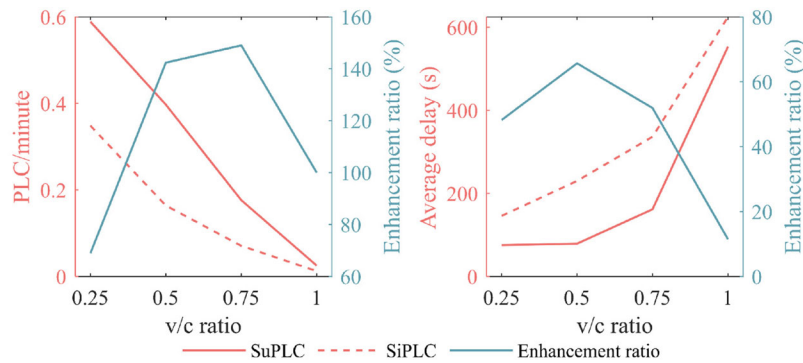


Figure 23. Sensitivity analysis of performance enhancement regarding v/c ratio.

- The proposed SuPLC controller is confirmed with string stability and lateral stability.
- The proposed SuPLC controller enhances PLC competence by 120.49% on freeways and 71.36% on arterials.
- The proposed SuPLC controller enhances PLC efficiency by 41.36% on freeways and 25.05% on arterials.
- The performance of the proposed SuPLC controller peaks when v/c ratio is between 0.5 and 0.75. This indicates that the proposed controller is more robust against congestion.
- The proposed SuPLC controller is ready for real-time application.
- In this research, the communication delay and actuation delay are not accommodated. Future research could consider enhancing the proposed controller accordingly.

Disclosure statement

No potential conflict of interest was reported by the authors.

Funding

This paper is partially supported by Shanghai Municipal Science and Technology Major Project 2021SHZDZX0100, Shanghai Oriental Scholar (2018), Tongji Zhongte Chair Professor Foundation (No. 000000375-2018082), and the Fundamental Research Funds for the Central Universities, National Natural Science Foundation of China (Grant No. 61903058).

References

- Bamieh, B., Jovanovic, M. R., Mitra, P., & Patterson, S. (2012). Coherence in large-scale networks: Dimension-dependent limitations of local feedback. *IEEE Transactions on Automatic Control*, 57(9), 2235–2249. <https://doi.org/10.1109/TAC.2012.2202052>
- Barth, M., & Boriboonsomsin, K. (2008). Real-world carbon dioxide impacts of traffic congestion. *Transportation Research Record: Journal of the Transportation Research Board*, 2058(1), 163–171. <https://doi.org/10.3141/2058-20>
- Bengtsson, H. H., Chen, L., Voronov, A., & Englund, C. (2015). *Interaction protocol for highway platoon merge* [Paper presentation]. 2015 IEEE 18th International Conference on Intelligent Transportation Systems.
- Bibeka, A., Songchitruksa, P., & Zhang, Y. (2021). Assessing environmental impacts of ad-hoc truck platooning on

- multilane freeways. *Journal of Intelligent Transportation Systems*, 25(3), 281–292. <https://doi.org/10.1080/15472450.2019.1608441>
- Campion, G., Bastin, G., & Dandrea-Novell, B. (1996). Structural properties and classification of kinematic and dynamic models of wheeled mobile robots. *IEEE Transactions on Robotics & Automation*, 12(1), 47–62. <https://doi.org/10.1109/70.481750>
- Cui, L., Hu, J., Park, B. B., & Bujanovic, P. (2018). Development of a simulation platform for safety impact analysis considering vehicle dynamics, sensor errors, and communication latencies: Assessing cooperative adaptive cruise control under cyber attack. *Transportation Research Part C: Emerging Technologies*, 97, 1–22. <https://doi.org/10.1016/j.trc.2018.10.005>
- Dang, D., Gao, F., & Hu, Q. (2020). Motion planning for autonomous vehicles considering longitudinal and lateral dynamics coupling. *Applied Sciences*, 10(9), 3180. <https://doi.org/10.3390/app10093180>
- Dey, K. C., Yan, L., Wang, X., Wang, Y., Shen, H., Chowdhury, M., Yu, L., Qiu, C., & Soundararaj, V. (2016). A review of communication, driver characteristics, and controls aspects of cooperative adaptive cruise control (CACC). *IEEE Transactions on Intelligent Transportation Systems*, 17(2), 491–509. <https://doi.org/10.1109/TITS.2015.2483063>
- Ding, T., Bo, R., Gu, W., & Sun, H. (2014). Big-M based MIQP method for economic dispatch with disjoint prohibited zones. *IEEE Transactions on Power Systems*, 29(2), 976–977. <https://doi.org/10.1109/TPWRS.2013.2287993>
- Hao, H., Barooah, P., & Mehta, P. G. (2011). Stability margin scaling laws for distributed formation control as a function of network structure. *IEEE Transactions on Automatic Control*, 56(4), 923–929. <https://doi.org/10.1109/TAC.2010.2103416>
- Hsu, H.-H., & Liu, A. (2004). Platoon lane change maneuvers for automated highway systems [Paper presentation]. IEEE Conference on Robotics, Automation and Mechatronics.
- Hsu, H. C.-H., & Liu, A. (2008). Kinematic design for platoon-lane-change maneuvers. *IEEE Transactions on Intelligent Transportation Systems*, 9(1), 185–190. <https://doi.org/10.1109/TITS.2007.908721>
- Hu, J., Wang, H., Feng, Y., & Li, X. (2022). Optimal control-based highway pilot motion planner with stochastic traffic consideration. *IEEE Intelligent Transportation Systems Magazine*, 2–17. <https://doi.org/10.1109/ITS.2022.3181172>
- Hu, J., Wang, H., Li, X., & Li, X. (2020). Modelling merging behaviour joining a cooperative adaptive cruise control platoon. *IET Intelligent Transport Systems*, 14(7), 693–701. <https://doi.org/10.1049/iet-its.2019.0378>
- Huang, Z., Chu, D., Wu, C., & He, Y. (2019). Path planning and cooperative control for automated vehicle platoon using hybrid automata. *IEEE Transactions on Intelligent Transportation Systems*, 20(3), 959–974. <https://doi.org/10.1109/TITS.2018.2841967>
- Kesting, A., Treiber, M., Schönhof, M., & Helbing, D. (2008). Adaptive cruise control design for active congestion avoidance. *Transportation Research Part C: Emerging Technologies*, 16(6), 668–683. <https://doi.org/10.1016/j.trc.2007.12.004>
- Khalil, H. K. (1994). Robust servomechanism output feedback controllers for feedback linearizable systems. *Automatica*, 30(10), 1587–1599. [https://doi.org/10.1016/0005-1098\(94\)90098-1](https://doi.org/10.1016/0005-1098(94)90098-1)
- Khound, P., Will, P., Tordeux, A., & Gronwald, F. (2021). Extending the adaptive time gap car-following model to enhance local and string stability for adaptive cruise control systems. *Journal of Intelligent Transportation Systems*, 1–21. <https://doi.org/10.1080/15472450.2021.1983810>
- Lai, J., Hu, J., Cui, L., Chen, Z., & Yang, X. (2020). A generic simulation platform for cooperative adaptive cruise control under partially connected and automated environment. *Transportation Research Part C: Emerging Technologies*, 121, 102874. <https://doi.org/10.1016/j.trc.2020.102874>
- Lam, S., & Katupitiya, J. (2013). *Cooperative autonomous platoon maneuvers on highways* [Paper presentation]. 2013 IEEE/ASME International Conference on Advanced Intelligent Mechatronics.
- Lawler, E. L., & Wood, D. E. (1966). Branch-and-bound methods: A survey. *Operations Research*, 14(4), 699–719. <https://doi.org/10.1287/opre.14.4.699>
- Li, L., Lu, Y., Wang, R., & Chen, J. (2017). A three-dimensional dynamics control framework of vehicle lateral stability and rollover prevention via active braking with MPC. *IEEE Transactions on Industrial Electronics*, 64(4), 3389–3401. <https://doi.org/10.1109/TIE.2016.2583400>
- Liu, H., Kan, X., Shladover, S. E., Lu, X.-Y., & Ferlis, R. E. (2018). Impact of cooperative adaptive cruise control on multilane freeway merge capacity. *Journal of Intelligent Transportation Systems*, 22(3), 263–275. <https://doi.org/10.1080/15472450.2018.1438275>
- Liu, H., Shladover, S. E., Lu, X.-Y., & Kan, X. (2021). Freeway vehicle fuel efficiency improvement via cooperative adaptive cruise control. *Journal of Intelligent Transportation Systems*, 25(6), 574–586. <https://doi.org/10.1080/15472450.2020.1720673>
- Margiotta, R., & Washburn, S. (2017). Simplified highway capacity calculation method for the highway performance monitoring system.
- Milanés, V., & Shladover, S. E. (2014). Modeling cooperative and autonomous adaptive cruise control dynamic responses using experimental data. *Transportation Research Part C: Emerging Technologies*, 48, 285–300. <https://doi.org/10.1016/j.trc.2014.09.001>
- National Center for Statistics and Analysis. (2019, June). Early Estimate of Motor Vehicle Traffic Fatalities in 2018.
- Nilsson, J., Brännström, M., Coelingh, E., & Fredriksson, J. (2015). *Longitudinal and lateral control for automated lane change maneuvers* [Paper presentation]. 2015 American Control Conference (ACC). <https://doi.org/10.1109/ACC.2015.7170929>
- Park, B. B., Malakorn, K., & Lee, J. (2011). *Quantifying benefits of cooperative adaptive cruise control towards sustainable transportation system*.
- Ploeg, J., Scheepers, B. T., Van Nunen, E., Van de Wouw, N., & Nijmeijer, H. (2011). *Design and experimental evaluation of cooperative adaptive cruise control* [Paper presentation]. 2011 14th International IEEE Conference on Intelligent Transportation Systems (ITSC).

- Ploeg, J., Van De Wouw, N., & Nijmeijer, H. (2014). Lp string stability of cascaded systems: Application to vehicle platooning. *IEEE Transactions on Control Systems Technology*, 22(2), 786–793. <https://doi.org/10.1109/TCST.2013.2258346>
- Qin, Y., & Wang, H. (2021). Stabilizing mixed cooperative adaptive cruise control traffic flow to balance capacity using car-following model. *Journal of Intelligent Transportation Systems*, 1–23. <https://doi.org/10.1080/15472450.2021.1985490>
- Qin, Y., Wang, H., & Ran, B. (2021). Impacts of cooperative adaptive cruise control platoons on emissions under traffic oscillation. *Journal of Intelligent Transportation Systems*, 25(4), 376–383. <https://doi.org/10.1080/15472450.2019.1702534>
- Saerens, B., Rakha, H., Ahn, K., & Van Den Bulck, E. (2013). Assessment of alternative polynomial fuel consumption models for use in intelligent transportation systems applications. *Journal of Intelligent Transportation Systems*, 17(4), 294–303. <https://doi.org/10.1080/15472450.2013.764801>
- Seiler, P., Pant, A., & Hedrick, K. (2004). Disturbance propagation in vehicle strings. *IEEE Transactions on Automatic Control*, 49(10), 1835–1842. <https://doi.org/10.1109/TAC.2004.835586>
- Shladover, S. E., Su, D., & Lu, X.-Y. (2012). Impacts of cooperative adaptive cruise control on freeway traffic flow. *Transportation Research Record: Journal of the Transportation Research Board*, 2324(1), 63–70. <https://doi.org/10.3141/2324-08>
- Swaroop, D., & Hedrick, J. K. (1996). String stability of interconnected systems. *IEEE Transactions on Automatic Control*, 41(3), 349–357. <https://doi.org/10.1109/9.486636>
- Toy, C., Leung, K., Alvarez, L., & Horowitz, R. (2002). Emergency vehicle maneuvers and control laws for automated highway systems. *IEEE Transactions on Intelligent Transportation Systems*, 3(2), 109–119. <https://doi.org/10.1109/TITS.2002.801422>
- Treiterer, J., & Myers, J. (1974). The hysteresis phenomenon in traffic flow. *Transportation & Traffic Theory*, 6, 13–38.
- Vander Werf, J., Shladover, S. E., Miller, M. A., & Kourjanskaia, N. (2002). Effects of adaptive cruise control systems on highway traffic flow capacity. *Transportation Research Record: Journal of the Transportation Research Board*, 1800(1), 78–84. <https://doi.org/10.3141/1800-10>
- Wang, H., Li, X., Zhang, Y., Sun, B., Ma, W., & Hu, J. (2019). Motion Planning Algorithm Under Partially Connected and Automated Environment.
- Wang, M., Daamen, W., Hoogendoorn, S. P., & van Arem, B. (2016). Connected variable speed limits control and car-following control with vehicle-infrastructure communication to resolve stop-and-go waves. *Journal of Intelligent Transportation Systems*, 20(6), 559–572. <https://doi.org/10.1080/15472450.2016.1157022>
- Weaver, S. M., Balk, S. A., & Philips, B. H. (2021). Merging into strings of cooperative-adaptive cruise-control vehicles. *Journal of Intelligent Transportation Systems*, 25(4), 401–411. <https://doi.org/10.1080/15472450.2020.1728264>
- Xu, Y., Ahmadian, M., & Sun, R. (2014). Improving vehicle lateral stability based on variable stiffness and damping suspension system via MR damper. *IEEE Transactions on Vehicular Technology*, 63(3), 1071–1078. <https://doi.org/10.1109/TVT.2013.2282824>
- Yang, M., Wang, X., & Quddus, M. (2019). Examining lane change gap acceptance, duration and impact using naturalistic driving data. *Transportation Research Part C: Emerging Technologies*, 104, 317–331. <https://doi.org/10.1016/j.trc.2019.05.024>
- Yu Zhang, Y. B., Wang, M., & Hu, J. (2020). Chang-Hu's Optimal Motion Planning Framework for Cooperative Automation: Mathematical Formulation, Solution, and Applications, Transportation Research Board Annual Meeting.
- Zhang, C., Chu, D., Liu, S., Deng, Z., Wu, C., & Su, X. (2019). Trajectory planning and tracking for autonomous vehicle based on state lattice and model predictive control. *IEEE Intelligent Transportation Systems Magazine*, 11(2), 29–40. <https://doi.org/10.1109/ITS.2019.2903536>
- Zhang, Y., Bai, Y., Wang, M., & Hu, J. (2021). Cooperative adaptive cruise control with robustness against communication delay: An approach in the space domain. *IEEE Transactions on Intelligent Transportation Systems*, 22(9), 5496–5507. <https://doi.org/10.1109/TITS.2020.2987746>

Appendices

Proof of proposition 1

The cost function of the proposed SuPLC controller in equation (19) is transformed into a continuous domain as follows:

$$\min_u \int_{s_r=0}^{s_r=S} L(\xi - \xi_{des}, u) ds_r \quad (79)$$

$$L = \frac{1}{2} (\xi - \xi_{des})^T Q (\xi - \xi_{des}) + \frac{1}{2} u^T R u \quad (80)$$

To adopt the PMP method, the state vector is set to $\xi - \xi_{des}$. The vehicle dynamics model in equation (15) is transformed into a continuous domain as follows:

$$\dot{f} = \frac{\partial(\xi - \xi_{des})}{\partial s_r} = A(\xi - \xi_{des}) + B u + C \quad (81)$$

(81)where

$$C^{Str} C^{Str}_a^0 - C^{Lat} \delta_f^0 \quad (82)$$

$$C^{Str} \left[(\pi^0(0))^3, 0, 0, 0 \right]^T \quad (83)$$

$$C^{Lat} \left[0, 0, \frac{1}{l_{fr}}, 0 \right]^T \quad (84)$$

Definition 3 is utilized to derive the control law. For a quadratic optimal control problem, the optimal value is where the derivative is zero. Therefore, equation (29) indicates:

$$\frac{\partial H}{\partial U^*} = 0 \quad (85)$$

By substituting equations (28), (80), and (81) into equations (85) and (31), the control law of the SuPLC controller is as follows:

$$u^* = -R^{-1} B^T \lambda \quad (86)$$

$$-\frac{d\lambda}{ds_r} = Q(\zeta - \zeta_{des}) + \mathbb{A}^T \lambda \quad (87)$$

By substituting equations (28), (34), and (35) into equations (85) and (31), the control law of the centralized platoon controller in Proposition 1 is as follows:

$$U_n^* = -R_n^{-1} \mathbb{B}_n^T \lambda_n \quad (88)$$

$$-\frac{d\lambda_n}{ds_r} = Q_n X_n + \mathbb{A}_n^T \lambda_n \quad (89)$$

where λ_n is the co-state vector of the centralized platoon controller in Proposition 1. λ_n is defined as follows:

$$\lambda_n \triangleq \begin{bmatrix} \lambda^1 \\ \lambda^2 \\ \vdots \\ \lambda^n \end{bmatrix}_{4n \times 1} \quad (90)$$

Substituting equations (36), (37), (38), (39), (40), (41), and (90) into equations (88) and (89):

$$u^j = -R^{-1} \mathbb{B}^T \lambda^j \quad (91)$$

$$-\frac{d\lambda^j}{ds_r} = Q(\zeta^j - \zeta_{des}^j) + \mathbb{A}^T \lambda^j \quad (92)$$

Equations (91) and (92) are the same with equations (86) and (87) for any follower j . Therefore, the control law of the SuPLC controller is the same as the control law of the centralized platoon control problem in Proposition 1. This platoon controller is equivalent to the SuPLC controller. This concludes the proof.

Proof of lemma 1

By applying Definition 4, state vector, control vector, and co-state vector are transformed into the frequency domain as follows:

$$X_n(s) \triangleq \mathcal{L}\{X_n(s_r)\} \quad (93)$$

$$U_n(s) \triangleq \mathcal{L}\{U_n(s_r)\} \quad (94)$$

$$r_n(s) \triangleq \mathcal{L}\{r_n(s_r)\} \quad (95)$$

$$\mu^j(s) \triangleq \mathcal{L}\{\mu^j(s_r)\} \quad (96)$$

By applying Definition 4 and equations (93), (94), (95), platoon dynamics function (equations (34)) and control law (equations (88), (89)) are transformed into the frequency domain as follows:

$$X_n = (sI - \mathbb{A}_n)^{-1} (\mathbb{B}_n U_n + \mathbb{C}_n) \quad (97)$$

$$U_n = -R_n^{-1} \mathbb{B}_n^T r_n \quad (98)$$

$$r_n = -(sI + \mathbb{A}_n^T)^{-1} Q_n X_n \quad (99)$$

Substituting equations (97) and (99) into equation (98):

$$U_n = (R_n - M_n \mathbb{B}_n)^{-1} M_n \mathbb{C}_n \quad (100)$$

$\mathbb{C}_n$ is longitudinal and lateral coupled as defined in equation (44). Following analysis proves no interaction exists between longitudinal and lateral perturbation.

Applying Proposition 1 and equation (96):

$$U_n = \begin{bmatrix} \mu^1 \\ \mu^2 \\ \vdots \\ \mu^n \end{bmatrix}_{2n \times 1} \quad (101)$$

$$\mu^j = (I - M\mathbb{B})^{-1} M\mathbb{C} \quad (102)$$

where

$$E \triangleq (sI + \mathbb{A}^T)^{-1} \quad (103)$$

$$F \triangleq (sI - \mathbb{A})^{-1} \quad (104)$$

$$M \triangleq \mathbb{B}^T E Q F \quad (105)$$

Substituting equations (20), (21), (42), and (43) into (103), (104), and (105):

$$E = \begin{bmatrix} s & 0 & 0 & 1 \\ 0 & s & 0 & 0 \\ 0 & 1 & s & 0 \\ 0 & 0 & 0 & s \end{bmatrix}^{-1} \quad (106)$$

$$F = \begin{bmatrix} s & 0 & 0 & 0 \\ 0 & s & -1 & 0 \\ 0 & 0 & s & 0 \\ -1 & 0 & 0 & s \end{bmatrix}^{-1} \quad (107)$$

$$M = \begin{bmatrix} 2q_1(\pi^0(0))^3 & 0 & 0 & sq_4(\pi^0(0))^3 \\ 0 & \frac{-sq_2}{l_{fr}} & \frac{-2q_3}{l_{fr}} & 0 \end{bmatrix} \quad (108)$$

Therefore:

$$a^j(s) = [1, 0](R - M\mathbb{B})^{-1} M \mathbb{C}^{Str} a^0(s) \quad (109)$$

$$\delta_f^j(s) = [0, 1](R - M\mathbb{B})^{-1} M \mathbb{C}^{Lat} \delta_f^0(s) \quad (110)$$

Equations (109) and (110) indicate that longitudinal and lateral behaviors are not coupled in the space domain. Therefore, by substituting equations (44), (53), (54), and (100) into equations (49) and (50), equations (51) and (52) are derived. This concludes the proof.

Proof of lemma 2

Given equations (38), (39), (40), (41), (44), (45), and (46):

$$Q_{n+1} = \begin{bmatrix} Q_1 & 0 \\ 0 & Q_n \end{bmatrix} \quad (111)$$

$$R_{n+1} = \begin{bmatrix} R_1 & 0 \\ 0 & R_n \end{bmatrix} \quad (112)$$

$$\mathbb{A}_{n+1} = \begin{bmatrix} \mathbb{A}_1 & 0 \\ 0 & \mathbb{A}_n \end{bmatrix} \quad (113)$$

$$\mathbb{B}_{n+1} = \begin{bmatrix} \mathbb{B}_1 & 0 \\ 0 & \mathbb{B}_n \end{bmatrix} \quad (114)$$

$$\mathbb{C}_{n+1}^{Str} = \begin{bmatrix} \mathbb{C}_1^{Str} \\ \mathbb{C}_n^{Str} \end{bmatrix} \quad (115)$$

$$\mathbb{C}_{n+1}^{Lat} = \begin{bmatrix} \mathbb{C}_{n+1}^{Lat} \\ \mathbb{C}_n^{Lat} \end{bmatrix} \quad (116)$$

Substituting equations (111), (112), (113), and (114) into equations (55), (56), and (57):

$$\mathbf{E}_{n+1} = \begin{bmatrix} \mathbf{E}_1 & 0 \\ 0 & \mathbf{E}_n \end{bmatrix} \quad (117)$$

$$\mathbf{F}_{n+1} = \begin{bmatrix} \mathbf{F}_1 & 0 \\ 0 & \mathbf{F}_n \end{bmatrix} \quad (118)$$

$$\mathbf{M}_{n+1} = \begin{bmatrix} \mathbf{M}_1 & 0 \\ 0 & \mathbf{M}_n \end{bmatrix} \quad (119)$$

Given equations (53) and (54):

$$\mathbf{O}_{n+1,j+1}^{Str} = \begin{bmatrix} 0 & \mathbf{O}_{n,j}^{Str} \end{bmatrix} \quad (120)$$

$$\mathbf{O}_{n+1,j+1}^{Lat} = \begin{bmatrix} 0 & \mathbf{O}_{n,j}^{Lat} \end{bmatrix} \quad (121)$$

Substituting equations (112), (115), (116), (117), (118), (119), (120), and (121) into equations (51) and (52):

$$\begin{aligned} \Gamma_{n+1,j+1,0}^{Str} &= \left\| \begin{bmatrix} 0 & \mathbf{O}_{n,j}^{Str} \end{bmatrix} \left(\begin{bmatrix} \mathbf{R}_1 & 0 \\ 0 & \mathbf{R}_n \end{bmatrix} - \begin{bmatrix} \mathbf{M}_1 & 0 \\ 0 & \mathbf{M}_n \end{bmatrix} \begin{bmatrix} \mathbb{B}_1 & 0 \\ 0 & \mathbb{B}_n \end{bmatrix} \right)^{-1} \right. \\ &\quad \left. \begin{bmatrix} \mathbf{M}_1 & 0 \\ 0 & \mathbf{M}_n \end{bmatrix} \begin{bmatrix} \mathbb{C}_{n+1}^{Str} \\ \mathbb{C}_n^{Str} \end{bmatrix} \right\|_2 = \left\| \mathbf{O}_{n,j}^{Str} (\mathbf{R}_n - \mathbf{M}_n \mathbb{B}_n)^{-1} \mathbf{M}_n \mathbb{C}_n^{Str} \right\|_2 = \Gamma_{n,j,0}^{Str} \end{aligned} \quad (122)$$

$$\begin{aligned} \Gamma_{n+1,j+1,0}^{Lat} &= \left\| \begin{bmatrix} 0 & \mathbf{O}_{n,j}^{Lat} \end{bmatrix} \left(\begin{bmatrix} \mathbf{R}_1 & 0 \\ 0 & \mathbf{R}_n \end{bmatrix} - \begin{bmatrix} \mathbf{M}_1 & 0 \\ 0 & \mathbf{M}_n \end{bmatrix} \begin{bmatrix} \mathbb{B}_1 & 0 \\ 0 & \mathbb{B}_n \end{bmatrix} \right)^{-1} \right. \\ &\quad \left. \begin{bmatrix} \mathbf{M}_1 & 0 \\ 0 & \mathbf{M}_n \end{bmatrix} \begin{bmatrix} \mathbb{C}_{n+1}^{Lat} \\ \mathbb{C}_n^{Lat} \end{bmatrix} \right\|_2 = \left\| \mathbf{O}_{n,j}^{Lat} (\mathbf{R}_n - \mathbf{M}_n \mathbb{B}_n)^{-1} \mathbf{M}_n \mathbb{C}_n^{Lat} \right\|_2 = \Gamma_{n,j,0}^{Lat} \end{aligned} \quad (123)$$

This concludes the proof.

Proof of lemma 3

Given equations (53) and (54):

$$\mathbf{O}_{n+1,1}^{Str} = \begin{bmatrix} \mathbf{O}_{1,1}^{Str} & 0 \end{bmatrix} \quad (124)$$

$$\mathbf{O}_{n+1,1}^{Lat} = \begin{bmatrix} \mathbf{O}_{1,1}^{Lat} & 0 \end{bmatrix} \quad (125)$$

Substituting equations (112), (115), (116), (117), (118), (119), (124), and (125) into equations (51) and (52):

$$\begin{aligned} \Gamma_{n+1,1,0}^{Str} &= \left\| \begin{bmatrix} \mathbf{O}_{1,1}^{Str} & 0 \end{bmatrix} \left(\begin{bmatrix} \mathbf{R}_1 & 0 \\ 0 & \mathbf{R}_n \end{bmatrix} - \begin{bmatrix} \mathbf{M}_1 & 0 \\ 0 & \mathbf{M}_n \end{bmatrix} \begin{bmatrix} \mathbb{B}_1 & 0 \\ 0 & \mathbb{B}_n \end{bmatrix} \right)^{-1} \right. \\ &\quad \left. \begin{bmatrix} \mathbf{M}_1 & 0 \\ 0 & \mathbf{M}_n \end{bmatrix} \begin{bmatrix} \mathbb{C}_{n+1}^{Str} \\ \mathbb{C}_n^{Str} \end{bmatrix} \right\|_2 = \left\| \mathbf{O}_{1,1}^{Str} (\mathbf{R}_1 - \mathbf{M}_1 \mathbb{B}_1)^{-1} \mathbf{M}_1 \mathbb{C}_1^{Str} \right\|_2 = \Gamma_{1,1,0}^{Str} \end{aligned} \quad (126)$$

$$\begin{aligned} \Gamma_{n+1,1,0}^{Lat} &= \left\| \begin{bmatrix} \mathbf{O}_{1,1}^{Lat} & 0 \end{bmatrix} \left(\begin{bmatrix} \mathbf{R}_1 & 0 \\ 0 & \mathbf{R}_n \end{bmatrix} - \begin{bmatrix} \mathbf{M}_1 & 0 \\ 0 & \mathbf{M}_n \end{bmatrix} \begin{bmatrix} \mathbb{B}_1 & 0 \\ 0 & \mathbb{B}_n \end{bmatrix} \right)^{-1} \right. \\ &\quad \left. \begin{bmatrix} \mathbf{M}_1 & 0 \\ 0 & \mathbf{M}_n \end{bmatrix} \begin{bmatrix} \mathbb{C}_{n+1}^{Lat} \\ \mathbb{C}_n^{Lat} \end{bmatrix} \right\|_2 = \left\| \mathbf{O}_{1,1}^{Lat} (\mathbf{R}_1 - \mathbf{M}_1 \mathbb{B}_1)^{-1} \mathbf{M}_1 \mathbb{C}_1^{Lat} \right\|_2 = \Gamma_{1,1,0}^{Lat} \end{aligned} \quad (127)$$

This concludes the proof.

Proof of lemma 4

Given Lemma 2,

$$\frac{\Gamma_{n+1,j+1,0}^{Str}}{\Gamma_{n,j,0}^{Str}} = \frac{\Gamma_{n+1,j,0}^{Str}}{\Gamma_{n,j-1,0}^{Str}} \quad (128)$$

$$\frac{\Gamma_{n+1,j+1,0}^{Lat}}{\Gamma_{n,j,0}^{Lat}} = \frac{\Gamma_{n+1,j,0}^{Lat}}{\Gamma_{n,j-1,0}^{Lat}} \quad (129)$$

Therefore,

$$\frac{\Gamma_{n+1,j+1,0}^{Str}}{\Gamma_{n+1,j,0}^{Str}} = \frac{\Gamma_{n,j,0}^{Str}}{\Gamma_{n,j-1,0}^{Str}} \quad (130)$$

$$\frac{\Gamma_{n+1,j+1,0}^{Lat}}{\Gamma_{n+1,j,0}^{Lat}} = \frac{\Gamma_{n,j,0}^{Lat}}{\Gamma_{n,j-1,0}^{Lat}} \quad (131)$$

Given equations (130) and (131), equations (62) and (63) are derived. This concludes the proof.